

Electronic Commerce, Fintech, and Trust Online

Modules 1 to 3 built the ground: what a system is, which one a company should build, and what all of it runs on. This module is about the moment money actually moves. Somebody you have never met, in a place you cannot see, sends you a payment for something you have not handed over yet — and both of you have to trust an arrangement neither of you built. Everything in this chapter is about how that trust is manufactured, what it costs, and who pays when it fails.

1

OBJECTIVE 4.1

Define electronic commerce and fintech, and say what part each plays in moving a business into digital trade

2

OBJECTIVE 4.5

Apply the key ideas of payment security and legal compliance to running an electronic commerce operation

3

APPLICATION 4.6

Name the platform, payment and fulfillment services a storefront actually sits on, and the technology answering back on the shop floor

4

APPLICATION 4.7

Turn all of it into the written analysis this course keeps asking for: pressure, value, one recommendation

This module is built differently

The earlier modules explain first and practise second, and they explain at length. This one is roughly a third of the reading and about twice the practice, because this material is judgement rather than vocabulary. Nobody struggles with the definition of a chargeback. They struggle with deciding whether to ship the order.

1. **Every section carries a mini case.** A short situation, a few facts, usually a small exhibit of figures, and two or three decisions to make. The debrief stays shut until you have answered all of them, so the point of the case is worked out rather than read.
2. **Read the short piece first, then work.** The prose in each section is deliberately brief and exists to arm the activity that follows it, not to substitute for it.

3. **Every option explains itself.** Including the ones you did not choose. Knowing why a plausible answer is wrong is the part that survives into the next case.
4. **Two supplements are labeled as such.** The last two sections are applications, not textbook objectives. They connect this chapter to the services a real storefront runs on and to the written analysis this course asks for.
5. **Finish with the final challenge.** Situations rather than definitions, scored by objective, so a weak area shows up instead of being averaged away.

Where this module sits in the course objectives

The modules follow the textbook's learning objectives, but the course is graded against a different list of eleven. This table says plainly which of them this module serves and which it leaves to other work, so you always know why you are reading something.

COURSE OBJECTIVE	WHAT THIS MODULE DOES WITH IT
5. Delineate how businesses use internet technology for electronic commerce, communicating with employees and working with business partners	This module's main job. Electronic commerce is the whole of objective 4.1; the business-to-business half is how firms work with partners; and the supplement on the commerce ecosystem is the platform, payment and fulfillment relationships a storefront actually runs on.
9. Recognize security breaches and computer crimes	Also central. Objective 4.5 is card-not-present fraud, stolen card data, account takeover, chargebacks and the screening that tries to catch all of it before an order ships.
3. Discuss how companies use information technology for competitive advantage by changing their marketplace	Also central. Cutting out the middle of a supply chain, selling straight to the customer, and answering an online competitor from a shop floor are all changes to a marketplace rather than improvements inside one.
8. Describe the individual and organizational consequences of the use of information technology	Substantial. Abandoned carts, the friction between security and convenience, restrictions placed on media people believe they bought, and the tracking that pays for a personalised store are all consequences landing on individuals.

COURSE OBJECTIVE	WHAT THIS MODULE DOES WITH IT
2. Discuss how companies use information systems to manage internal operations and make decisions	Supporting. Fraud screening produces a score somebody has to act on, and the fulfillment decision is an operations decision made from data.
7. Recognize a business problem which might be solved using information technology, then work with the information systems team on a solution	Practised in the last supplement, which is a full pass from evidence to one recommended initiative.
1. Describe what a company's information systems department does · 4. Apply a working knowledge of hardware, software, database technology and networks	Carried by Modules 1 and 3. This module assumes them rather than repeating them.
10. Solve a business problem by creating a database application using SQLite · 11. Use advanced spreadsheet functions to create a decision support system	Prepared for, not completed, in the last supplement: a runnable sheet and a runnable query put this chapter's questions to a storefront's own order data. Designing a database and building an application on it belong to the course's own database and spreadsheet work.
6. Outline the life cycle of an information system	Not taught here.

REACH

Selling without a shop

Electronic commerce moved the transaction onto a network, and in doing so let a producer reach a customer directly — removing whoever used to stand between them.

MONEY

Paying without a bank branch

Fintech did the same thing to financial services: one job, done very well, reaching the customer without the institution that used to sit in the middle.

RISK

Trusting a stranger

No card to hold, no signature, no face. The merchant carries the loss when the transaction is disputed, so screening an order is a business decision, not a technical one.

RULES

Trading across borders

A single online storefront can fall under consumer protection, tax and intellectual property rules from places nobody at the company has ever been.

ONE IDEA CONNECTS THE WHOLE MODULE

Every arrangement in this chapter exists to replace a form of trust that used to be physical. A shop you can walk into, a card you can hold, a signature, a bank with a door on the street. Take those away and something has to stand in for them — a payment service, a public ledger, a risk score, a returns policy, a law. Each of those substitutes works, and each one costs somebody something. Most of the decisions in this module are about which cost to accept.

SECTION 4-1A

What electronic commerce actually is

Most people meet electronic commerce as a shopping site. The definition is wider and older than that, and the wider version explains why a business changes shape when it starts selling online.

It started with the data, not the shopping

In 1948 the United States Air Force in Europe was flying supplies into Berlin around the clock. The cargo mattered, and so did the information about it: what a shipment held, where it was, and when it would land.

To move that information the operation devised standard universal codes and sent them by teletype. Transactions between organizations became electronic there, in a records problem, decades before anybody bought anything from a web page.

The modern definition keeps that shape. **Electronic commerce** is electronic transactions over digital networks: the exchange of goods, services and money among firms, between firms and their customers, and between customers. Four parts of that sentence each do work.

Electronic transactions — the agreement, the money and the record all move as data rather than as paper handed across a counter.

Over digital networks — the two parties never have to be in the same place, or awake at the same hour, for the exchange to complete.

Goods, services and money — what is exchanged may be a physical object, something delivered entirely as data, or the payment itself.

Among firms, with customers, between customers — all three directions are inside the definition, so one private seller trading with one private buyer counts.

More than the moment of payment

Notice what the definition does not say: it does not say buying. Electronic commerce covers what leads up to a purchase and the service that follows it: a buyer who cannot find a specification never reaches the payment step. Sort the nine moments below into the part of the arc each belongs to.

SORT

Where in the arc does each moment sit?

Place each moment in the part of the arc it belongs to; the buyers and sellers described are generic examples rather than real firms.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Before the sale — everything that helps a buyer decide, up to the moment of commitment

- Comparing the specifications of two models side by side on a seller's site — Comparison is where most purchases are won or lost, and the definition covers it: electronic commerce includes the events leading up to a purchase, not only the purchase.
- An email reminding a shopper about the basket they left without paying — The basket was abandoned, so no sale has happened. The reminder is an attempt to finish a purchase that is still ahead of the buyer rather than behind them.
- A price quotation sent to a business buyer who asked what 200 units would cost — A quotation is an offer, not a sale. It is one of the pre-purchase events the definition covers, and a great deal of selling between organizations happens at this step.

The transaction itself — the exchange of commitment and money, and the record that closes it

- The card network authorizing the amount as the buyer pays — Authorization is the moment the money is committed. It is the narrow event most people mean by electronic commerce, and it is the smallest part of the whole arc.
- The order confirmation the buyer receives seconds later — The confirmation is the record that closes the exchange. The boundary is genuinely close here, so read it as the receipt belonging to the transaction rather than as later service.

After the sale — everything the seller still owes the buyer once the money has moved

- A notice that the parcel has left the depot and will arrive on Thursday — Delivery tracking is service after the sale. The money has already moved, and the seller is now keeping a promise rather than trying to win one.
- A printable label for returning an item that did not fit — Returns are the clearest post-sale obligation, and an awkward one is a common reason a buyer does not come back, which is why sellers treat returns as part of commerce.
- A registration page for the two-year warranty on a repaired item — Warranty registration happens once the item is owned, and it exists to serve the buyer later on, which places it firmly on the far side of the sale.
- A review left three weeks after delivery — The review is written long after the money moved, and it then feeds the comparison step of the next buyer, which is how the arc loops back to its own beginning.

Cutting out the middleman

The internet and the web then widened all of that, and selling to individual consumers became practical. The web changed how firms create and capture value: a producer that once needed somebody else to reach buyers could now reach them itself, which broke business models that had been stable for a century.

Disintermediation is the name for that: cutting out the middleman — the wholesalers and retailers that traditionally sat between a producer and its customers — and reaching customers directly. Airlines selling seats on their own sites are one example, and direct-to-consumer brands are another.

Selling direct is not the only outcome. Many producers who leave a wholesaler behind list inside an **electronic marketplace** instead — a shared venue such as Amazon or Etsy, where many sellers reach many buyers and an almost limitless range of services can be offered. It supplies the reach and charges for it.

Compare the three arrangements below, and watch who is carrying the cost of reaching the customer in each one.

COMPARE

Three routes from producer to customer

Step through the three arrangements and read the notes under each; the roles are generic rather than any particular company.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The chain before the web — Each step buys from the one before it, adds a margin, and sells on.

Producer (Makes the product) → Wholesaler (Buys in bulk, holds stock) → Retailer (Sells in ones and twos) → Customer (Known only to the retailer)

- Each intermediary is paid for something real: reach, storage, credit and the work of selling in small quantities.
- The producer's own selling cost is low, because the two middlemen carry the cost of reaching the customer.
- The producer learns almost nothing about the buyer, since every conversation happens two steps away from it.
- Margin is lost at each handover, which is the figure that makes selling direct look attractive.

Disintermediated — The producer sells straight to the customer over its own site.

Producer (Makes, markets, ships, supports) → Customer (Buys from the producer itself)

- Disintermediation removes the intermediaries and their margin, and hands the producer everything they were doing.
- The cost of reaching the customer now sits with the producer, as advertising, parcels, returns and support.
- In exchange the producer gains the relationship and the data, which the longer chain never allowed it.
- The chapter's examples are airlines selling seats on their own sites and the rise of direct-to-consumer brands.

Re-intermediated — A different intermediary takes the place of the old one.

Producer (Makes and ships the product) → Online marketplace (Finds the buyer, takes a cut) → Customer (Buys inside the marketplace)

- The marketplace performs the old intermediary's function — reach and trust — and charges a fee for it.
- The producer still packs and ships, but the buyer arrives through somebody else's front door.
- Removing one middleman often installs another, because the work of finding buyers does not vanish with the wholesaler.
- Whether this is the better deal depends on the fee set against what reaching those buyers alone would cost.

Whichever arrangement a producer ends up in, the middleman's work does not disappear along with the middleman. It moves, and a distributor usually does more than its invoice suggests.

Reach — it already sells to hundreds of shops, so a product arrives in front of buyers the producer has never met.

Stock and delivery — it holds goods, consolidates orders and ships in bulk, which costs far less per unit than posting single parcels.

Money and risk — it pays on invoice and carries the loss when a shop pays late or not at all, so the producer is not financing the sale.

Returns and questions — it absorbs damaged goods, replacements and the enquiries that follow a sale, none of which reach the producer at all.

Every one of those is work, and disintermediation hands it to the producer along with the margin. The case below is a hypothetical workshop weighing exactly that move, with invented figures.

MINI CASE

Sell direct, or keep the distributor?

Read the brief and the exhibit, then take the three decisions in order; the workshop is invented for this exercise and every figure in it was made up for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A two-person workshop makes one product, a hand-built bicycle pannier, and sells every unit to a single distributor that resells to bike shops. A developer has offered to build the workshop an online store of its own. The owners have to decide whether to sell direct as well, and they have one quarter of figures to reason from. The workshop is hypothetical.

The product

One item, hand-built to order in small batches

Route to market today

Every unit goes to one distributor, which resells to bike shops

What the distributor absorbs

Stock, bulk shipping, invoicing, late payment, damaged goods, shop enquiries

Who the workshop hears from

The distributor only; it has never spoken to a rider

The decision

Open a direct online store alongside the distributor, or leave things as they are

Exhibit A • One pannier, two routes — Per-unit economics for a single quarter, side by side. Every figure here is invented for practice and none of it is reported data.

LINE	THROUGH THE DISTRIBUTOR	SELLING DIRECT
Price the workshop receives	US\$48	US\$95

LINE	THROUGH THE DISTRIBUTOR	SELLING DIRECT
Packing and shipping	US\$3 on a shared pallet	US\$11 per parcel
Payment handling fee	None; paid on invoice	US\$2.85 per order
Returns and replacements	Absorbed by the distributor	US\$4 averaged per unit
Answering buyer questions	Absorbed by the shops	US\$6 of staff time per unit
Units sold last quarter	410	None; the store does not exist yet

1. Which comparison on the exhibit actually decides this, rather than merely looking decisive?

- **A.** The two prices received, since the direct figure is nearly double the other one — The headline price is the most visible figure here and the least informative, because the distributor column pays for work that the direct column has to buy separately.
- **B.** The price received minus every cost listed in the same column, for both columns — **correct.** Two lines in the distributor column read as absorbed rather than as zero, so the routes only become comparable once every cost in each column is taken off the price that column earns.
- **C.** The packing and shipping line, since it is the cost that rises the most — Packing and shipping does rise, from a share of a pallet to single parcels, but it is one line among several and settles nothing on its own.
- **D.** The payment handling fee, since it is a cost the workshop does not pay today — The handling fee is genuinely new to the workshop, and it is also the smallest of the direct costs, so treating it as the deciding figure misreads the exhibit.

2. The owners agree the fully costed margin is better direct. What does disintermediating transfer to the workshop along with that margin?

- **A.** The distributor's work as well — stock, parcels, chasing payment, returns, questions — **correct.** Every line the exhibit marks as absorbed is a task somebody still has to perform, and after the change that somebody is the workshop, which is most of why the margin is larger.
- **B.** The distributor's existing relationships with the bike shops that stock the product — Shop relationships belong to the distributor and do not travel. Selling direct reaches a different buyer entirely, and the shops may well treat the new store as competition.
- **C.** The risk of damage in transit, which now sits with the shipping carrier instead — A carrier does take on some transit risk, but the workshop still replaces the damaged pannier and still answers the rider who received it, whatever the carrier eventually pays.

- **D.** The distributor's buying data, which transfers with the account when it closes — The distributor has never shared its buying data, and an ended contract does not oblige it to. That data is precisely what the workshop has been missing all along.

3. Set the money aside. What does the direct route give the workshop that it has never had before?

- **A.** A lower cost of getting a single pannier into a buyer's hands than today — Shipping single parcels costs the workshop more per unit than a share of a pallet did, so the cost of one delivery moves in the wrong direction here.
- **B.** Contact with the riders themselves, and a record of what they ask for — **correct.** The facts note that the workshop has never spoken to a rider; selling direct produces both the relationship and a record of what buyers want, and no distributor invoice contains either.
- **C.** Freedom from the fees that come with taking payment online — Taking payment online adds a fee rather than removing one, and the exhibit prices it at a small but real amount on every direct order the workshop takes.
- **D.** An assurance that the total volume sold each quarter will rise — Nothing here promises more units. The last row of the exhibit says the direct volume is untested, which is the largest unknown in the whole decision.

What this case was really about. This is disintermediation in miniature. Removing the middleman moves its margin to the producer and its work along with it, so an honest comparison prices both columns fully and then asks whether the direct volume even exists. What the workshop gains that no distributor ever gave it is the customer relationship and the data attached to it, which is the deeper reason an online presence became a strategic necessity rather than a cost saving.

Why an online presence stopped being optional

Two properties explain why an online presence became a strategic necessity for most companies, and neither of them is about fashion.

Market reach — a site is open to anyone who can reach it, so the market is no longer only the people who can travel to a building.

Scalability — serving the thousandth visitor costs far less than serving the first, so growth need not mean a matching increase in premises and staff.

Neither property says the online route is cheaper to run. It is a different way to reach a market, and it arrives carrying all of the work the earlier list described.

Electronic commerce is not the payment. It is every electronic step from the first comparison to the last replacement, and disintermediation moves all of those steps — not only the margin — onto whoever now stands closest to the customer.

Three questions to test the definition, the term and the arc before you move on.

CHECK YOURSELF

Definition, disintermediation, and the whole arc

Four options, one best answer; read every explanation, including those for the options you did not choose, and treat any company described as hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. The chapter dates the foundations of electronic transactions between businesses to a 1948 airlift. What is the point of starting there?

- **A.** It was the first time goods were ordered and paid for without the two parties meeting — Buying at a distance long predates 1948. Letters, telegrams and travelling agents had done it for centuries without any of it being electronic in the sense used here.
- **B.** The information about the cargo mattered as much as the cargo itself — **correct.** The operation realised that what was in a shipment, where it had got to and when it would arrive mattered as much as the shipment, and devised standard universal codes to send those data by teletype.
- **C.** The military built the first computer network, which businesses later borrowed for trade — The airlift used teletype rather than a computer network. The internet arrived decades later and widened electronic commerce rather than founding it.
- **D.** Air freight made long-distance trade practical, which is what commerce had been waiting for — Air freight matters to logistics, but the chapter is making a point about the data describing the cargo, not about the speed at which the cargo itself moved.

2. A regional airline closes its agency accounts and sells every seat through its own site. Which description fits what has happened?

- **A.** Re-intermediation: a different intermediary has taken the place of the old one — Re-intermediation is what happens when a producer lists on a large marketplace instead. Here nobody new has taken up position between the airline and the passenger.
- **B.** An electronic marketplace: many sellers now trade seats inside one shared venue — An electronic marketplace hosts many sellers in one venue. The airline's own site offers only its own seats, which is a narrower arrangement than that.
- **C.** Disintermediation: the intermediary is gone and its work moves to the airline — **correct.** Cutting out the traditional intermediary and reaching customers directly is disintermediation, and the chapter names airlines selling direct as a prime example; the agents' work now belongs to the airline.

- **D.** Scalability: the airline can serve more passengers without adding more staff — Scalability is one reason an online presence is attractive rather than a name for this change, and serving more passengers is not what removing the agents did.

3. A furniture maker's site publishes specifications, takes orders, tracks deliveries and issues return labels. The owner says only the payment step counts as electronic commerce. Why is that too narrow?

- **A.** Because the electronic record is the test, so tracking is inside it and returns are not — The site does record the tracking, but what the site logs is not the boundary. A return label the maker issues sits inside the definition just as firmly as the tracking page does.
- **B.** Because a site with several functions counts as an electronic marketplace instead — An electronic marketplace is a venue where many sellers offer to many buyers. One maker running one site does not become one by adding functions to it.
- **C.** Because the payment step is run by a payment service rather than by the maker — Who operates the payment step is a separate question, and it would not narrow the definition even if an outside service handled every transaction on the site.
- **D.** Because the definition also covers what leads up to a purchase and what follows it — **correct.** Electronic commerce includes the events leading up to a purchase and the customer service after the sale, so the specifications, the tracking and the return labels are all inside it.

Who is on each side of the transaction

Electronic commerce is usually sorted into types, and the sorting rule is simpler than it looks: name who stands on each side of the exchange. Not the website, not the product, not the size of the order. This section works through the types the chapter names, and through the cases where the obvious answer is the wrong one.

Naming the two sides

The convention is a pair of letters with a “to” between them. The first letter is the side that sells or provides. The second is the side that buys or receives. The chapter names three types built on that pattern.

Business-to-consumer (B2C) — transactions between businesses and their customers, which covers online retailing, the sale of digital products, and streaming services.

Business-to-business (B2B) — transactions among businesses, such as a manufacturer trading over the web with its suppliers, with no end consumer in the exchange.

Consumer-to-consumer (C2C) — transactions between people who are not necessarily working together, such as one person buying memorabilia from another through an auction site.

When neither side is a business

Some exchanges have no firm on either side. One person sells to another person, both of them are consumers, and the transaction is C2C.

The marketplace hosting them is doing something else. It is not the seller. It runs a **platform business model**: it enables transactions between other parties and takes a fee for that service. The goods on offer are not its goods.

A fourth category sits outside all three, because it involves services neither sold by nor to companies. **E-government** is the use of information systems to give citizens, organizations and other government agencies information about public services, and to let them interact with the government.

MATCH**What actually distinguishes each one**

Match each term to the thing that really separates it from its neighbors, then read why the boundary sits where it does.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Business-to-consumer (B2C)

A firm on the selling side and an end consumer on the buying side This is the part of electronic commerce most people can see, which is why it gets mistaken for the whole of it. Online retailing, digital products and streaming services all sit here.

Business-to-business (B2B)

An organization on both sides, with no end consumer involved The chapter calls this by far the largest form by revenue, and it is invisible to shoppers. A manufacturer trading over the web with its suppliers is the standard case.

Consumer-to-consumer (C2C)

A person on each side, not necessarily working together The test is the two parties, not the venue. An auction sale between two individuals stays consumer to consumer however large and commercial the site hosting it is.

E-government

Information about public services, and interaction with government This one is defined by what a public body provides rather than by who buys from whom. A government purchasing software from a vendor is trading business to business instead.

Platform business model

Enabling other parties to transact, and earning a fee for it This names the host's role rather than the type of transaction. A marketplace can run a platform under consumer-to-consumer sales without owning any of the goods on it.

Notice what the naming rule does not use. It says nothing about the size of the order, the technology behind the checkout, or whether the seller owns a warehouse. Two sides, named. That is the whole test.

The chapter's e-government examples are worth reading slowly, because each one replaces a counter, a queue or a paper form.

Internet tax filing — the most recognizable service of the group, and one that saves resources in both time and paper.

Identity card applications — the application is submitted online rather than in person at a counter during opening hours.

Municipal services — the everyday business of a city, requested and tracked through the city's own systems.

Electronic voting — casting a ballot through a system rather than on paper, where a jurisdiction has chosen to offer it.

Public data access — electronic access to government data that anybody may read and use for their own purposes.

Services to organizations — filings and reporting offered online to businesses and to other government agencies.

The chapter's own summary table lists three types. The fourth row below is e-government, which the chapter defines in the paragraph beside that table rather than inside it.

TYPE	DESCRIPTION	EXAMPLE
Business-to-consumer (B2C)	Transactions between businesses and their customers	A person buys a book from an online retailer.
Business-to-business (B2B)	Transactions among businesses	A manufacturer trades over the web with its suppliers.
Consumer-to-consumer (C2C)	Transactions between people not necessarily working together	A person buys memorabilia from another person through an auction site.
E-government	Information about public services, and interaction with government	A resident files a tax return through a government system.

Those rows are easy to read and easy to misapply, which is what the sorting exercise is for. Every organization named in it is invented.

SORT Who is on each side here?

Place each transaction under the type it belongs to; every organization described is a hypothetical example.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Business to consumer — a firm on the selling side, an end consumer on the buying side

- A student pays the monthly charge for a streaming subscription. — Streaming services are one of the chapter's own examples of business-to-consumer commerce: a firm on one side, an end consumer on the other.
- A shopper buys a novel as a download rather than as a printed copy. — The sale of digital products is business to consumer for the same reason a physical delivery is. What changes is the shipping, not who the two parties are.
- A parent orders school shoes from a retailer's website for home delivery. — This is the case everybody pictures first, and it is worth keeping as a reference point: a business selling to the person who will actually use the goods.

Business to business — an organization on both sides, with no end consumer in the exchange

- A bakery orders forty sacks of flour through its mill's ordering site. — Both sides are firms and the flour goes into something the bakery will sell, so no end consumer appears anywhere in this exchange.
- An office manager buys twelve desk chairs from a well-known consumer retailer's website. — The site was designed for shoppers, but the type follows the parties rather than the design. A firm buying chairs for its own operations is on the business side.
- A city buys a licence for the software that runs its permit desk. — A public body appears here, which tempts people toward e-government. E-government names the services a government provides, not the purchases it makes from vendors.
- A manufacturer's suppliers sign in to a portal to confirm delivery dates. — This is the chapter's own illustration of business-to-business commerce, and it shows that the category covers the events around a purchase, not just the payment.

Consumer to consumer — a person on each side, whoever hosts the page they meet on

- Someone sells a used bicycle to a stranger through an auction site that charges a fee. — The fee makes the host a business, but the host is not a party to the sale. Two consumers are, so the transaction is consumer to consumer.
- A retiree sells garden tools to a neighbor through a local listings site. — Neither side is a firm and the two are not working together, which is exactly how the chapter describes consumer-to-consumer commerce.

E-government — public services and interaction with government, delivered by information system

- A resident renews a driving licence on a state website. — A citizen is interacting with government about a public service, which is the definition of e-government rather than a purchase of goods.
- A household files its tax return through the tax authority's online system. — Internet tax filing is the chapter's most recognizable e-government example, and it saves resources in both time and paper.

- An agency publishes its spending records for anyone to download. — Providing electronic access to public government data is listed by the chapter as e-government, even though nobody buys or sells anything at all.

Where the money actually is

Most people meet electronic commerce as B2C, because that is the part of it they can see. By revenue it is the smaller half, and not by a little. B2B is the largest form of all, because a great many firms trade with other firms and never with a consumer.

The chapter reports 2022 figures for United States manufacturers' electronic-commerce shipments and for wholesalers' electronic-commerce sales, each measured in trillions of dollars. Learn the ordering rather than the digits: business-to-business volume dwarfs the consumer-facing volume, and a figure like that is a snapshot of one year.

The second thing to carry away is that one firm is rarely one type. Take a consumer-facing company — the chapter's example is a clothing and home furnishing retailer — and follow a single product through it.

- 1 It **sources raw materials** from a range of specialized suppliers, and every one of those purchases is business to business.
- 2 It **sells the finished product** to a distributor or a wholesaler, which is business to business a second time.
- 3 Or it **sells that product directly** to the end consumer, and only this last step is business to consumer.

The shopper sees the last step and assumes it is the business. The steps before it were larger, and they were the other type. The case below is built on that mismatch, inside an invented company.

MINI CASE Four revenue streams, one website

Read the brief and the exhibit, then take the three decisions in order; the company and every figure in it are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Ridgehollow Roasterworks is a hypothetical company used here for practice. It builds coffee-roasting equipment and earns money four ways through one website: a shop for home roasters, contract supply to cafe chains, a resale area where owners sell used machines to each other, and an equipment contract with a community college district's kitchens. The founder wants to know which of the four the company actually runs on, and whether one website can serve them all.

Online shop

Home roasters buy a single machine and accessories directly.

Cafe contracts

Chains order production roasters under a supply agreement.

Resale area

Owners list used machines to each other; the company takes a fee.

College kitchens

A community college district buys and restocks equipment.

One storefront

All four streams run through the same catalog and checkout.

Exhibit A • one year of trading — The four streams side by side. Every figure here is invented for practice and none of it is reported data.

REVENUE STREAM	ORDERS IN THE YEAR	AVERAGE ORDER VALUE	REVENUE TO THE YEAR
Online shop, home roasters	4,120	US\$310	US\$1,277,200
Contract supply, cafe chains	24	US\$152,000	US\$3,648,000
Resale area, owner to owner	1,450	US\$240	US\$52,200 in fee
College kitchens, district contract	30	US\$8,100	US\$243,000

1. The resale area carries the company's branding and sits one click from the shop, so shoppers treat it as part of the store. Which type are those transactions, and what is the company doing there?

- **A.** C2C, with the company running a platform that enables the sale and charges a fee — **correct.** Both parties are individuals, so the exchange is consumer to consumer, and the company's role is the platform business model: it enables the transaction and earns a fee for doing so.
- **B.** B2C, because the used machines reach ordinary consumers through the company's own storefront and branding — The branding belongs to the company, but the seller does not. A consumer on each side makes this consumer to consumer, whoever designed the page it happens on.
- **C.** B2B, because the sellers are equipment owners rather than ordinary shoppers — Business to business needs an organization on both sides. Two machine owners trading with each other are consumers here, however much equipment they happen to own.

- **D.** Not electronic commerce at all, since the company sells nothing of its own there — Money moves over a digital network in exchange for goods, which is electronic commerce by the chapter's definition, and the fee the company keeps is real revenue.

2. Look at the exhibit as a whole. Which stream carries the most revenue, and how does that sit beside what the chapter reports about the types?

- **A.** The online shop, since it wrote by far the largest number of orders across the year — Order count and revenue are different measurements. The shop wrote by far the most orders and still earned roughly a third of what the cafe contracts brought in.
- **B.** The cafe-chain contracts, on the fewest orders of the four — **correct.** Two dozen contract orders outearn every other stream combined, which is the chapter's finding in miniature: business-to-business trade is the largest form by revenue.
- **C.** The resale area, because it moved more machines than either contract stream did — The resale area moved a great many machines, but the company keeps only a listing fee from each one, which leaves it the smallest of the four streams.
- **D.** The college kitchens, since public buyers commit to the largest yearly totals — The college contract is real business-to-business revenue and it is steady, yet the exhibit puts it at a small fraction of what the cafe chains spent.

3. The founder plans one identical storefront for all four streams: the same catalog, the same checkout, the same card payment at the end. What does that decision cost the company most?

- **A.** Nothing that can be measured, since every buyer wants a fast checkout above everything else — Speed matters to a home buyer choosing one machine. A chain buying two dozen roasters a year is working to a budget cycle and an approval process instead.
- **B.** The resale area, which cannot function without a card payment at its checkout — The resale area does need a payment path, and card payment suits it well, because the amounts involved are close to ordinary retail purchases.
- **C.** The college contract, which is too small a share of revenue to design around — Share of revenue is not the same as fit. The district buys on a purchase order much as the chains do, so one design decision serves or fails both of them.
- **D.** The stream earning the most, whose buyers work by purchase order — **correct.** The cafe contracts earn the most and behave the least like retail: negotiated prices, purchase orders and invoices rather than a card typed into a checkout.

What this case was really about. Four streams, one company, and four different answers — while the website stayed the same in every one of them. The type is decided by who stands on each side: two organizations, an organization and a consumer, or two consumers with a platform in between. The stream that looked least like a shop carried most of the money, which is the chapter's point about business-to-business trade at the scale of one firm.

Two boundaries do most of the damage. A consumer-facing brand can be on the buying side of a business-to-business purchase, and a marketplace can host a consumer-to-consumer sale without ever being a party to it.

Ask who is on each side before you ask anything else. The type is a fact about the two parties, not about the storefront they happened to meet on.

CHECK YOURSELF

Reading the two sides correctly

Four options, one best answer; read every explanation, including the ones for options you did not choose. Any organization described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. How do business-to-consumer and business-to-business commerce compare in revenue, according to the chapter?

- **A.** B2C is larger, because far more people shop online than there are businesses to trade with — Shopper numbers are not revenue. A great many firms trade with other firms and never with a consumer, and their orders are far larger than a household's.
- **B.** The two are close, since most firms sell to consumers and to businesses alike — Many consumer-facing firms do trade both ways, which is true and useful, but it says nothing about the relative size of the two totals.
- **C.** B2B is the larger of the two by a wide margin — **correct.** The chapter treats business-to-business commerce as by far the largest form by revenue, and reports United States shipment and wholesale figures in the trillions for 2022.
- **D.** The chapter offers no comparison, because the two are measured on different bases — The chapter does compare them, and gives figures for a single year to do it. Those figures are a snapshot, which is a reason to learn the ordering rather than the digits.

2. A county buys accounting software from a vendor and, in the same week, opens a site where residents can pay property bills. Which types are these two things?

- **A.** Both are e-government, because a public body is one of the parties in each of them — The presence of a government body does not settle the type. E-government names the services a government provides to citizens and organizations, not every purchase it makes.
- **B.** The software purchase is business to business; the payment site is e-government — **correct.** Buying software puts an organization on each side, which is business to business, while the payment site provides a public service and interaction with government.
- **C.** The software purchase is e-government; the payment site is business to consumer — This reverses both halves. A purchase from a vendor is trade between organizations, and residents paying a public bill are interacting with government rather than shopping.

- **D.** Neither is electronic commerce, because no physical goods change hands here — The chapter's definition covers the exchange of goods, services and money over digital networks, and software and payments are services traded exactly that way.

3. A marketplace lets private owners list secondhand goods to other private buyers, charges its sellers a fee, holds the buyer's money until delivery, and never owns any of the goods. How should its transactions and its own role be described?

- **A.** C2C transactions, with the marketplace running a platform business model — **correct.** Both parties to the sale are individuals, so the exchange is consumer to consumer, and the host earns a fee for enabling it rather than for selling anything.
- **B.** B2C transactions, since the marketplace is the party collecting payment from the buyer — Collecting the payment is a service the marketplace performs. The goods belong to the seller throughout, and that seller is another consumer rather than a firm.
- **C.** B2B transactions, because the sellers are paying fees to the marketplace as a business — A fee paid by a seller is a business arrangement with the host, but the transaction being classified is the sale of goods between two people.
- **D.** C2C transactions in which the marketplace is one of the two parties — A party to the sale would own the goods and carry the risk attached to them. Enabling the sale and holding the money briefly is a different role entirely.

Fintech: money without the middle

Buying something online is only half a transaction. The other half is the money: how it leaves one account, how long it takes to arrive, and who takes a slice on the way. This section is about the firms that rebuilt that half, and about what they did not rebuild.

One service, done very well

Fintech is the chapter's word for innovative technologies that transform financial services. In plainer terms it is software doing a job a bank used to do. The chapter credits it with accelerating electronic commerce three ways: simpler online payments, easier financing for online sellers, and new products such as digital wallets and peer-to-peer payments.

One structural difference explains most of what follows. A traditional bank offers a wide range of services under one roof — an account, a card, a loan, somewhere to invest. A fintech firm typically takes one of those and does it very well.

Depth in one service lets a young firm compete with an institution a century old. Narrowness obliges its customers to hold several relationships instead of one. The chapter's own categories all follow that shape.

Crowdfunding platforms — many small backers fund one project directly, instead of one lender deciding on everybody's behalf.

Online-only banks — everyday accounts, cards and transfers run entirely through an app, with no branch anywhere behind them.

Mobile payment processors — a phone or a small reader becomes a way for almost any seller to accept cards.

Peer-to-peer lenders — people with money to lend are matched against people who want to borrow it.

Small-business lenders — credit for firms too small or too quick-moving for a traditional commercial loan process.

Digital investment managers — a portfolio built and rebalanced by software from a short questionnaire about goals.

EXPLORE**Six narrow firms, four questions each**

Open each card and read all four panels; the same four questions are asked of every category, which is what makes them comparable, and no real company is named.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Crowdfunding platforms · Many small backers, one project

- **What it does:** Lets a project raise money from a large number of small contributors, each deciding for themselves, instead of from one institution deciding on behalf of many.
- **Who it displaces:** The lender or investor who would otherwise have assessed the project and committed other people's money to it, along with the meetings that assessment required.
- **What it does not solve:** It does not judge whether the project can actually be delivered. Backers carry that risk themselves, and a funded project is not a finished one.
- **Why it can be cheaper:** The platform runs one listing process and one payment flow for every campaign, so the cost of administering a single deal is spread across all of them.

Online-only banks · An account with no branch

- **What it does:** Provides everyday banking — holding money, moving it, and a card to spend it with — through an app, with no counter anywhere to walk up to.
- **Who it displaces:** The branch network, and with it the premises, the staff and the local opening hours that used to be how a customer reached their own money.
- **What it does not solve:** Anything that still wants a physical presence: paying in cash, or a conversation with somebody who has the authority to make an exception for you.
- **Why it can be cheaper:** Premises and branch staff are among a traditional bank's largest fixed costs, and an app serves one more customer at close to no extra cost.

Mobile payment processors · Card acceptance for a stall

- **What it does:** Turns an ordinary phone or a small reader into a way of accepting card payments, so a seller can take a card without a contract negotiated in advance.
- **Who it displaces:** The bank arrangement and the fixed terminal that once stood between a small seller and the ability to accept anything other than cash.
- **What it does not solve:** The fee on each sale, which is often above a large retailer's negotiated rate; the convenience is paid for out of a slice of every transaction.
- **Why it can be cheaper:** Signing up is automated and the hardware is inexpensive, so the processor spends almost nothing acquiring a seller who may take only a few payments a week.

Peer-to-peer lenders · Savers lending to borrowers

- **What it does:** Matches people who have money to lend against people who want to borrow it and prices the loan, taking a fee for running the match rather than lending its own funds.

- **Who it displaces:** The bank in the middle, which traditionally took deposits at one rate, lent that money out at a higher one, and kept the difference.
- **What it does not solve:** Default. If a borrower stops paying, the loss falls on the people who funded that particular loan rather than on an institution absorbing it.
- **Why it can be cheaper:** There is no branch, no deposit base to fund and a credit decision made largely by software, so the gap between what a lender earns and a borrower pays can be narrower.

Small-business lenders · Credit judged on trading data

- **What it does:** Lends to small firms, often deciding from trading records such as card takings and sales history rather than from a long written application and a meeting.
- **Who it displaces:** The commercial loan officer, and the weeks of assessment a small borrower used to sit through for an amount the bank considered marginal.
- **What it does not solve:** The cost of borrowing for a risky business. A fast decision is not a cheap one, and speed is frequently priced into the rate.
- **Why it can be cheaper:** The assessment runs automatically against data the business already generates, so a small loan stops being uneconomic to underwrite.

Digital investment managers · A portfolio run by software

- **What it does:** Builds and rebalances an investment portfolio from a few answers about goals and tolerance for loss, and increasingly uses artificial intelligence to advise or to manage it.
- **Who it displaces:** The human adviser whose time made small portfolios uneconomic to serve, together with the annual review appointment that came with the relationship.
- **What it does not solve:** Market risk, and anything about the customer that the questionnaire did not think to ask about in the first place.
- **Why it can be cheaper:** One model serves every account at once, so the extra cost of managing one more portfolio is close to nothing.

A seventh belongs beside those six: artificial intelligence giving investment advice or managing a portfolio, which applies the same move to judgement rather than to paperwork. Firm names are left out on purpose. Companies of this kind are bought, merged and renamed often, and the category outlives the brand.

Cutting out the middle, again

Section 4–1a introduced **disintermediation**: reaching customers directly by removing the middle, as an airline does by selling a seat on its own site rather than through an agent. Fintech is that same move, aimed at financial intermediaries instead of wholesalers and retailers.

An online-only bank has no branch network standing between it and the customer. A direct investing app has no broker taking the order. The intermediary is not talked into charging less. It is taken out of

the path, and what replaces it costs almost nothing to run one more time.

So electronic commerce is no longer only trade. It once meant firms buying from firms and consumers buying from firms. It now takes in the financial transaction itself — banking, brokerage, lending, investment — and most major banks and investment firms provide those services online as well.

Expectations do not stay where they were set

Fintech improves the experience of paying, which is the visible effect. The quieter one is that it resets what a customer counts as normal. Once money moves between two people in seconds, a two-day wait somewhere else stops reading as ordinary and starts reading as a fault.

Expectations set in one industry are then applied to every other. Somebody who splits a restaurant bill instantly on a phone brings that standard to an insurer, a landlord and a payroll department, none of which ever competed with a payments app and all of which are now measured against one.

COMPLETE IT

The five load-bearing sentences

Choose the wording that matches the chapter in each sentence; the explanation appears once you pick, and it says what the other choices actually describe.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- The chapter's term for innovative technologies transforming financial services is **fintech**, and its effect on electronic commerce was to simplify online payments, streamline financing for sellers, and add products such as digital wallets. Fintech is the chapter's own word, and it names the technologies rather than any one company. E-government is the use of information systems to give citizens and organizations access to public services, which is a different form of electronic commerce. The electronic marketplace is the space where offerings meet buyers, not the technology reshaping financial services.
- Whereas a traditional bank provides a wide range of services, a fintech company typically **focuses on one service and does it very well**. That single sentence is the structural difference the section is built on, and it explains both the depth these firms reach and the number of relationships their customers end up holding. Matching a bank service for service would make the firm a bank, which is the opposite of the pattern. Selling through the incumbent describes a supplier relationship rather than a competitor bypassing one.
- Just as the internet let firms bypass wholesalers and retailers, fintech firms bypass traditional financial intermediaries and reach consumers directly. That move is **disintermediation**, the idea section 4–1a applied to goods, now applied to money. Disintermediation is cutting out the middle and reaching the customer directly, which is exactly what an online-only bank or a direct investing app does. Social commerce is selling through social platforms, an area electronic commerce spread into rather than a change in the chain. Consumer-to-consumer trade describes who the two parties are, not whether an intermediary was removed.

- Electronic commerce today reaches beyond business-to-business and business-to-consumer trade to take in **banking, brokerage, lending and investment**, with most major banks and investment firms providing those services online. The chapter widens the definition to the financial transaction itself, which is why fintech sits inside this objective at all. Shipping and warehousing are logistics that support a sale rather than transactions conducted online in their own right. Advertising and research are activities around the sale, and the chapter treats them as events leading up to a purchase.
- Fintech services enhance the customer experience and also **reshape what customers expect on ease and speed**, which is why a wait that felt ordinary a few years ago can now read as a fault. The chapter names ease of payment and speed as the expectations these services shape, and an expectation set in one industry is then carried into every other. Volumes have risen rather than fallen as paying became easier, so the second choice reverses the effect. Regulation is untouched by any of this: who may hold money and on what terms is decided elsewhere.

A bank sells breadth. A fintech firm sells depth. Almost every argument in this section is really about which of the two a particular customer needs at a particular moment.

Four money jobs in one small business

Strip a small trading business down and its money handling is four separate jobs. They feel like one job because a single bank account has always covered all four of them.

Taking payment in person — a card tapped at a stall, which needs a reader, an account to settle into and a fee per sale.

Taking payment online — a checkout that works on a phone and accepts money from somebody the seller will never meet.

Financing stock — borrowing now to buy in bulk against sales that will not arrive for weeks, judged on the business rather than the owner.

Parking a surplus — putting cash that is not needed this month somewhere it earns a return and stays reachable.

For each of those four there is now a specialist that does only that job. Splitting them buys a better price or a faster settlement on at least one. It also buys four logins, four fee schedules and four sets of records that somebody has to reconcile into one picture.

The questions underneath do not move either. Whoever holds the money is regulated by somebody, a balance is protected under some scheme or under none, and a lending decision made by software is still a lending decision. The business below is invented and its figures with it.

MINI CASE**Four jobs, one bank account, one evening to decide**

Read the brief and Exhibit A, then take the three decisions in order; the business is invented for practice and every figure in the exhibit is invented with it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Wrenfield Honey is a hypothetical two-person business that packs honey from local beekeepers and sells it at weekend markets and through a small website. Every payment it takes and every bill it pays runs through one ordinary business bank account. Trade is growing, a bulk purchase of this year's honey falls due in eight weeks, and about US\$4,000 sits idle most months. The pair have been told to look at specialist services, and they have one evening to decide what to change.

The business

Two people, hypothetical, packing and selling honey

Money job one

Taking card payments in person at weekend markets

Money job two

Taking card payments through the website

Money job three

Financing a bulk purchase of honey

Money job four

Parking a cash surplus of about US\$4,000

Today

All four run through a single business bank account

Exhibit A · the bank against a specialist, job by job — Every figure here is invented for practice and none is quoted from a real provider; each job appears twice, once as the bank offers it and once as a specialist offers it.

JOB AND PROVIDER	COST	WHEN THE MONEY ARRIVES	WHAT IT DOES NOT DO
In person · the bank	US\$25 a month, plus 1.9% of each sale	Three working days	Supply a reader that runs away from a power socket

JOB AND PROVIDER	COST	WHEN THE MONEY ARRIVES	WHAT IT DOES NOT DO
In person · specialist	No monthly fee, plus 2.6% of each sale	Next working day	Help much when a customer disputes a sale
Online · the bank	US\$40 a month, plus 1.8% of each sale	Three working days	Give a checkout page the site can simply drop in
Online · specialist	No monthly fee, plus 2.9% of each sale	Two working days	Cover a card used without its owner's consent
Honey purchase · the bank	9% a year, decision in about ten weeks	On drawdown after signing	Reach a decision before the supplier's deadline
Honey purchase · specialist	14% a year, decision in two days	Two days after approval	Reprice the loan once trading improves

JOB AND PROVIDER	COST	WHEN THE MONEY ARRIVES	WHAT IT DOES NOT DO
Surplus · the bank	Pays 0.1% a year on the balance	Available the same day	Earn a return worth the paperwork
Surplus · specialist	Pays 3.4% a year on the balance	Two working days to withdraw	Sit under the same protections as a bank balance

1. Reading Exhibit A, which of the four jobs does the specialist clearly win, and on what grounds?

- **A.** The surplus, because a much higher advertised rate on a balance is a plain gain — The rate on the idle balance is genuinely better and probably worth taking, but 3.4% against 0.1% on about US\$4,000 comes to roughly US\$130 across a whole year, and the last column raises a protection question the bank balance does not have.
- **B.** In-person cards, because a smaller share of each sale is taken by the specialist — Read the cost column again. The specialist takes 2.6% of each sale against the bank's 1.9%, so it is dearer per sale; what it removes is the monthly fee, which is a different argument entirely.
- **C.** Financing, because the bank's decision lands after the supplier's deadline — **correct.** The bank is cheaper on rate and useless on timing: a decision in about ten weeks cannot fund a purchase due in eight, so the dearer option is the only one that does the job at all.
- **D.** Online cards, because the money reaches the account on the same working day — Exhibit A shows two working days against the bank's three. That is an improvement of one day, which is worth having and is nowhere near the difference between a purchase happening and not happening.

2. The pair decide to move all four jobs to four different specialists. What have they taken on that the single account did not carry?

- **A.** Four sets of records that must be reconciled into one view of the business — **correct.** Fintech firms are narrow by design, so four strong services means four logins, four fee schedules and four statements, and in a two-person business somebody spends an evening making them agree with each other.
- **B.** Higher charges on all four of the jobs at once, since specialist pricing runs above a bank's — Exhibit A does not show that. The specialist is dearer per sale and dearer on the loan, and cheaper on monthly fees and better on the return paid on cash, so the direction differs from job to job.

- **C.** A regulator for the first time, because specialist providers are supervised — Financial providers of this kind already operate under supervision, so no new exposure appears here. What does need checking is which scheme, if any, protects a balance that is held outside a bank account.
- **D.** The loss of card acceptance at the weekend markets where they trade — Nothing about the move stops them taking cards at a market, and the in-person specialist exists precisely to make card acceptance easy for a stall. The cost of the move is administrative rather than operational.

3. One of the pair says the move means the business no longer has to think about any of this. Which reply is right?

- **A.** Automation across the four services means that the monthly bookkeeping now takes care of itself — Each service will report its own activity cleanly enough. None of them sees the other three, so the single view of the business that a lender or a tax return needs still has to be assembled by somebody or by a tool bought for the purpose.
- **B.** The providers take on the tax duties along with the payments they process — A payment processor moves money and reports what it moved. What is owed, to whom and where follows from the business and the places it sells into, and that obligation stays with the owners throughout.
- **C.** The narrower each provider is, the fewer records the business has to keep — Narrow providers keep excellent records of their own slice and none at all of anybody else's, so the number of records the business has to hold goes up when four services replace one account.
- **D.** Reconciling the four services and the rules about who holds the money are still theirs — **correct.** Fintech changed where each job is done and how fast the money moves. It did not remove the reconciling, and it did not answer whether any scheme protects a balance held outside a bank, which is a question for somebody qualified in this jurisdiction.

What this case was really about. This is disintermediation with a price on it. On three jobs the bank is adequate and the specialist buys a small improvement; on the fourth the bank's product does not exist inside the time available, and that is where a narrow firm earns its place. What splitting costs is coherence: four depth relationships instead of one breadth relationship, with the reconciling and the regulatory questions left exactly where they always were.

Three last checks on the vocabulary before the chapter turns from moving money to protecting it.

CHECK YOURSELF

Breadth, depth, and who got removed

Four options, one best answer; read every explanation, including those for the options you did not choose, and note that any business described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A customer compares a large bank with a firm that does nothing but process card payments. Which difference is the one the chapter draws between them?

- **A.** The bank is supervised by regulators, while a payments specialist answers to nobody at all — Both are supervised, and the chapter draws no such line. Regulation varies by activity and jurisdiction, which is a real question for a business, but it is not what separates a bank from a narrow firm here.
- **B.** The bank covers a wide range of services; the specialist covers one and does it well — **correct.** Breadth against depth is the structural difference the chapter states, and nearly everything else about fintech follows from it, including why customers end up holding several relationships.
- **C.** The specialist works with businesses, while the bank looks after private individuals — Plenty of specialist firms serve consumers directly, and banks serve businesses of every size. Who the customer is does not distinguish the two, and the chapter names categories on both sides of that line.
- **D.** The specialist is a smaller bank, offering the same range of services to fewer customers — A narrow firm is not a miniature bank. It aims to serve very large numbers of customers with a single service done well, so what marks it out is the range it offers rather than the size of its customer base.

2. An investing app lets a customer buy shares directly from a phone, with no broker taking the order. Which idea from this chapter does that illustrate?

- **A.** Consumer-to-consumer commerce, because the trade happens between two private individuals — Consumer-to-consumer trade describes people transacting with each other, as on an online auction site. The customer here is dealing through a firm, and the counterparty is a market rather than a neighbour.
- **B.** Digital investment management, since software rather than an adviser picks the holdings — That category covers a portfolio a service builds and rebalances for the customer from a short questionnaire. Here the customer chooses the trade, and what the app takes out of the path is the broker.
- **C.** Business-to-business commerce, since the app and the exchange are both firms — The customer is an individual buying for themselves, which makes this business-to-consumer. Firms are certainly involved in the plumbing, but the transaction the question describes ends with a consumer.
- **D.** Disintermediation, since a step in the chain is removed rather than made cheaper — **correct.** The chapter presents fintech as disintermediation applied to money: an intermediary is taken out of the path, exactly as an airline selling its own seats removes a travel agent.

3. A furniture retailer's customers have begun complaining that its refunds take three working days, though its policy has not changed in years. What best explains the complaints?

- **A.** Expectations set by instant payment services are now applied to every other industry — **correct.** The chapter notes that fintech services shape expectations on ease and speed, and those expectations do not stay in the industry that set them; the retailer is being measured against a payments app.
- **B.** The retailer has been disintermediated by a manufacturer selling directly to those customers — Disintermediation would show up as lost sales to a direct seller, not as complaints about the speed of a refund from a purchase the retailer did make. Nothing here says a competitor removed it from the chain.
- **C.** Refunds of this kind became a consumer-protection matter for the first time this year — Consumer protection in electronic commerce is long established rather than new, and a rule change would alter the policy or the deadline itself rather than produce complaints about an unchanged practice.
- **D.** Three working days is unusually slow by the standards of card refunds generally — Card refunds commonly take a few days to appear, so this figure is not unusual within its own industry. The point is that customers are comparing it with a faster industry instead of with its peers.

Paying online without handing over everything

Every online sale ends the same way: a stranger is asked to type a card number into a page they have never seen before. This section is about that moment — why it is where a sale is won or lost, and how a payment service changes who ends up holding the dangerous part.

Handing the sensitive part to somebody else

Concern about that moment produced independent **payment services** — PayPal, Apple Pay and Square among them — which stand between the shopper and the seller. The shopper pays using an account held with the service instead of typing card details into the seller's page.

The consequence is structural rather than cosmetic, and it is worth setting out item by item.

The card number — goes to the payment service and stops there, so it never reaches the seller's order system in the first place.

The account and its password — belong to the service, which means the seller is not defending a login that guards somebody's money.

The email address and purchase history — are kept by the service alongside the payment details, and are not shared with the merchant.

What is left with the seller — is an order, a name and an address: what it needs to ship the parcel, and nothing beyond that.

Turn that around and the point becomes plain. The merchant stops holding the thing a thief would want. The order still exists and the shipment still happens; the card number is simply somewhere else.

The sequence below is the ordinary checkout most people have already used, read for what each step takes off the seller's hands.

SEQUENCE

What happens when a shopper pays through a payment service

Put the six steps into the order they actually happen, then read what each one removes from the seller's exposure.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Nothing in this sequence is unusual; it is the checkout most people have used without thinking about it. The point of putting it in order is to notice what the seller never receives.

1. **At checkout the shopper chooses the payment service instead of the form that asks for a card number.** — Nothing sensitive has been typed into the seller's page, so there is nothing on that page to intercept and nothing the seller can store by accident.
2. **The shopper is handed over to the payment service and signs in to the account they already hold with it.** — The credential being checked belongs to the service. The seller never sees the password and never has to build or defend a login that guards anybody's money.
3. **Inside the service, the shopper sees who is being paid and how much, and approves the payment.** — The approval happens where the payment details already live, so the seller receives a decision rather than the card number that produced it.
4. **The service charges the card or account it holds for that shopper and moves the money to the seller.** — This is the step that would otherwise have put a card number into the seller's system. Here the number stays with the service, along with the email address and purchase history it keeps.
5. **The service tells the seller the payment succeeded and returns the shopper to the shop's confirmation page.** — What crosses to the seller is a confirmation and an order reference. Somebody who later breaks into the seller's records finds an order list rather than a set of card numbers.
6. **The seller ships the order, keeping the details it needs to deliver it and to answer questions about it.** — A name and an address stay with the seller, because no parcel can be posted without them. The part a thief would have wanted was removed four steps earlier and never arrives.

The moment the sale is lost

Keeping transactions secure is one of the most important parts of selling to consumers, of consumers trading with each other, and of any purchase made on a phone. Money moves at that moment, and so does everything needed to move it.

The chapter is blunt about the human half of the problem. People are reluctant to change their habits online, and they hand sensitive information to sites they know nothing about. As a snapshot of what that costs, the chapter reports that identity fraud cost Americans about US\$23 billion in 2023.

Shoppers who feel that unease respond by leaving. The chapter reports that more than half of online shopping carts are abandoned, and it names four reasons sitting behind most of it.

Security concerns — the shopper is not sure this seller can be trusted with a card number, and walking away from a cart costs them nothing at all.

Impatience — the purchase was a passing intention rather than a plan, and every extra second gives that intention time to fade.

Lengthy checkout procedures — each additional field and page is one more chance to decide the item was not worth this much effort.

Comparison shopping — the cart was being used as somewhere to park a candidate while the same item was priced somewhere else.

Read those four together and one management idea follows, and the rest of this objective turns on it. A seller has to strike a balance between providing security and reducing friction. Tighten the checkout and more shoppers leave; loosen it and the seller carries more risk.

CHECK YOURSELF

Friction, and who holds what

Four options, one best answer; read every explanation, including the ones for options you did not choose. Any shop described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A shop adds two more fields and one more page to its checkout so it can learn more about its buyers. Judging by the chapter's account of abandonment, what should it expect?

- **A.** More carts abandoned, since a lengthy checkout is one listed reason shoppers quit — **correct.** The chapter names lengthy checkout procedures beside security concerns, impatience and comparison shopping as reasons more than half of carts are abandoned.
- **B.** No change, because a shopper who reaches the checkout has already decided to buy — Reaching a checkout is an intention rather than a commitment. The chapter's point about abandonment is precisely that shoppers leave after they have started, which is how the figure passes half.
- **C.** Fewer abandoned carts, because the added detail makes the shop look more established — Looking established does help, but it comes from clear policies and a recognisable way to pay rather than from the number of questions asked before the money moves.
- **D.** A fall in fraud losses that comfortably outweighs anything lost at the checkout — Screening for fraud does draw on data collected at checkout, and that trade is real enough to be worth its own section later. It does not change what extra fields do to finished orders.

2. A shopper pays a small online seller through a payment service. Which description matches what the seller ends up holding?

- **A.** The card number and the address, both encrypted by the payment service beforehand — The service does not pass the card number along in any form. The design is that the sensitive information stops there rather than travelling onward in a protected wrapper.
- **B.** The order and the delivery details, but not the card number the shopper used — **correct.** The shopper gives the payment information only to the service, which keeps it along with details such as the email address and purchase history, and does not share it with the merchant.
- **C.** Nothing about the buyer at all, since the service handles the whole relationship — The seller still needs a name and an address to post the parcel, so it is not blind to the buyer. What it gives up is the payment detail, not the customer.
- **D.** A copy of everything the service holds, kept in case the service is unavailable — No such copy is made. Duplicating the service's records would recreate exactly the store of card data that using the service was meant to remove.

3. Tapping a phone to pay for a coffee, and splitting a restaurant bill with friends in an app, share something with paying an online shop through a payment service. What is it?

- **A.** The payment credentials stay with the service; the other party gets only a transfer — **correct.** In each case the payment details sit with the service, and the shop, the coffee counter or the friend receives money and a confirmation rather than the details behind it.
- **B.** The money moves without any company keeping a record of who paid whom — Records certainly are kept. The service knows the parties, the amount and the time, which is part of how a dispute can be settled later; what changes is who holds them.
- **C.** The transfer costs nothing, because no card network is involved at any point in it — Fees vary and are often invisible to the payer, but they are not absent. Payment services charge for what they do, which is a cost any shop adopting one has to weigh.
- **D.** Both parties must hold accounts at the same bank before the payment can clear — Peer-to-peer services are built to work across different banks and cards, which is what makes splitting a bill practical when nobody at the table banks in the same place.

A decision about payment security is never only a security decision. Every step you add to protect a sale is a step some shoppers will not finish, so the same choice shows up again in the revenue figures.

A payment service is not free. It charges a fee, and the fee lands on every completed order rather than only on the ones that would otherwise have gone wrong. The shop in the case below is invented, and so are its figures; the trade it faces is not.

MINI CASE**The step the owner was not looking at**

Read the brief, the facts and the exhibit, then take the three decisions in order; the shop, its funnel and every figure in it are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A hypothetical online shop sells specialist walking and climbing gear. Plenty of shoppers fill a basket and few of them finish. The owner is convinced the delivery charge is driving them away and wants to cut it. Before that, an adviser pulls one week of checkout data and asks whether a payment service, which charges a fee on every completed order, would be the better change.

What it sells

Specialist walking and climbing gear, online only

Shoppers reaching the basket

About 1,000 a week

Average order

US\$96, at roughly 45 percent gross margin

Where card details sit today

Stored in the shop's own order system

What the owner believes

That shoppers leave because delivery costs too much

Payment service fee, if adopted

US\$2.60 on each completed order

Exhibit A · One week of checkouts — Of every 1,000 shoppers who reach the basket, how many are still there after each step. Every figure here is invented for practice.

CHECKOUT STEP	STILL THERE, OF 1,000	LOST AT THIS STEP
1. Basket	1,000	—
2. Create an account and sign in	610	390
3. Delivery address	540	70
4. Delivery choice and cost	500	40
5. Card details	320	180
6. Confirm and pay	290	30

1. The owner is about to cut the delivery charge. Which line of the exhibit should change that plan?

- **A.** Step 2, where 390 shoppers leave rather than create an account — **correct.** The account requirement costs 390 of the 1,000, more than steps 3 to 6 lose between them, and it is demanded before the shopper has any reason to trust this shop.
- **B.** Step 4, where the delivery cost is shown for the first time — The delivery step loses 40 shoppers, the second smallest loss on the sheet. The owner's explanation is the one line the numbers do not support at all.
- **C.** Step 5, where 180 shoppers leave rather than type a card number — The card step is the second largest loss and is worth fixing, but it costs 180 shoppers against the 390 lost one screen earlier. Deal with the larger one first.
- **D.** Step 6, where 30 shoppers stop at the final confirmation — Thirty shoppers stopping at the confirmation is ordinary. By that point they have spent effort and made the decision, so it is the wrong place to look for the missing sales.

2. The payment service would let shoppers sign in with their payment account and skip the card screen, and the owner expects completions to rise from 290 to about 360 per 1,000. He objects to paying a fee on every order. What settles the question?

- **A.** The fee falls on completed orders the shop would have won anyway, so it should be refused — The fee does land on orders the shop would have won anyway, which is the honest part of the objection. It is not a reason on its own, because what matters is the net result rather than whether every dollar of fee was strictly needed.
- **B.** Compare the fee across all completed orders with the margin the extra orders bring in — **correct.** The fee on 360 orders is about US\$936 a week, while the 70 additional orders add roughly US\$6,720 in sales and about US\$3,024 in margin. The change pays for itself several times over.
- **C.** Nothing can be decided until the shop knows what its current card processor charges — Knowing the current processing cost would sharpen the figure and is worth finding out. It does not decide this one, because the gap between the fee and the recovered margin is far too wide for it to close.
- **D.** The fee is paid whether or not an order completes, so treat it as a fixed monthly cost — The facts strip says the fee is charged on completed orders, so it behaves like a cost of sale rather than rent. Reading it as a fixed cost makes a good change look worse than it is.

3. With the service in place the shop no longer stores card numbers in its own order system. In a year with no fraud at all, what has that bought?

- **A.** Nothing that year; the benefit only appears in a year when the shop is actually breached — A breach is not the only thing this changes, and waiting for one is the reasoning that leaves the data sitting there in the meantime. The exposure exists on every ordinary day, whether or not anybody has come for it yet.
- **B.** A smaller pile of data for anyone who does get in, which is worth something daily — **correct.** The chapter's point is structural: the payment information sits with the service, so what the shop holds is an order rather than a card number. That is true on quiet days too.

- **C.** A transfer of chargeback costs onto the payment service, which is why the fee exists — Chargebacks are a genuine cost for online merchants and a later section deals with them properly. A payment service does not simply absorb them, and that is not what this fee buys.
- **D.** The same records held in a safer form, which is what the shop pays the service for — Nothing here says the shop keeps an encrypted copy of its own. The change is that the card details never reach the shop at all, which is a different and stronger thing than storing them carefully.

What this case was really about. The case did two jobs. It located the friction, which was an account demanded before any trust existed rather than the delivery charge the owner blamed. And it separated two benefits that arrive together: fewer steps, so more orders finish, and less sensitive data held, so a bad day costs less. That pair is the balance between security and friction, priced.

Where these services went next

Google linked its payment service to its search results, so that somebody searching for a product can see immediately whether a merchant offers that way of paying. The stated intention was to ease the shopping experience and, by doing so, to reduce the number of abandoned carts.

PayPal went in a different direction, letting anyone with an email address send and receive money rather than only pay a shop. One person could now pay another, which was instrumental in the rise of the auction marketplace eBay, where buyer and seller are both ordinary people.

The chapter describes one of those two as owned by the other. That ownership ended in 2015 and they are separate companies now, so what matters here is the shared history: a payment tool and a marketplace that each made the other workable.

Phones then extended the same idea in two directions at once.

- Paying inside a shop by holding a phone to a terminal, which is an alternative to carrying cash or cards.
- Paying a person rather than a business, so a table of friends can settle a restaurant bill between them.

The Chinese messaging platform WeChat shows how far that can run: online payments, in-person payments and transfers between people, all inside a service people already open for other reasons.

The balance is the whole objective in one sentence. A payment service is attractive because it moves both sides of it at once: fewer steps for the shopper to abandon, and less sensitive data left sitting with the seller.

READY?**Could you make this call yourself?**

Rate each statement honestly; anything you cannot do yet has a pointer to the exact part of this section to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can say roughly how many online carts are abandoned and name the four reasons the chapter gives for it. Go back to the part on the moment the sale is lost, where the four reasons are listed one by one.
- I can state the balance a seller has to strike, and explain why it is a revenue decision as well as a security one. Go back to the balance paragraph in the part on the moment the sale is lost, and to the takeaway that follows it.
- I can describe what a payment service is and say exactly what the merchant stops holding once one is in place. Go back to the part on handing the sensitive part to somebody else, and to the fourth step of the sequence.
- I can walk through a checkout paid by a payment service and say what each step removes from the seller's exposure. Go back to the six-step sequence, which you can reorder as many times as you like.
- I can read a checkout funnel and find where shoppers are actually leaving rather than where I assume they are. Go back to the mini case, especially its first decision and the exhibit that decision turns on.

Cryptocurrency, blockchain, and trust without a middleman

Every online payment you have ever made passed through somebody you were trusting to make it right. This section takes that arrangement apart: what a trusted middleman actually sells, what it charges for the service, and what a payment looks like when there is nobody in the middle at all.

What the fee is buying

When you give a card number to a familiar retailer, or pay a small seller through a payment service, you are relying on a company you have never met to stand behind the transaction. The chapter calls those companies trusted middlemen. The trust is the product.

Two things in particular are being bought, and it helps to separate them.

A safety net — confidence that the money you send will produce the good or service you were promised, and a route to a refund when it does not.

Confidentiality — your financial information stays with the middleman rather than being handed to every seller you buy from, which is the whole reason payment services exist.

Neither comes free. The chapter puts card processing at roughly one to three percent of every purchase, charged to the vendor and normally passed on to everybody in higher prices. Payment services charge as well, sometimes several percentage points of the total.

A charge like that disappears inside a large purchase and dominates a small one. The case below works out where that line falls for a hypothetical firm sending many small payments to many countries.

MINI CASE

Four hundred small payments, thirty countries

Read the brief and the exhibit, then make the three decisions; the agency, the translators and every figure in the exhibit are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A hypothetical translation agency pays about four hundred freelance translators once a month. They live in roughly thirty countries, and most invoices are small. The finance lead has been asked whether the middleman is worth what it costs, and what the agency would be giving up if it stopped paying one. Nothing here is drawn from a real company.

Payment run

About 400 translators, once every month

Countries reached

Roughly 30

Typical invoice

About US\$18, with a long tail under US\$8

Largest invoices

A handful above US\$600

Disputes last year

Nine, all settled by the payment service

Exhibit A · the cost of one payment, four ways — The same US\$18 payment costed by method, with the fixed part and the percentage part shown separately. Every figure is invented for practice and is not quoted from any provider.

METHOD	FIXED PART	PERCENTAGE PART	COST ON US\$18	WHAT THE PAYEE GETS
Card processing	US\$0.30	2.5%	US\$0.75	Little; the right to survive
Payment service	US\$0.35	3.0%	US\$0.89	Little; the service provider disputes
International bank transfer	US\$14.00	none	US\$14.00	Speed; several days to land
Ledger-based transfer	US\$0.05	none	US\$0.05	Reversal and any process

1. Set the four rows of the exhibit against the agency's invoice sizes. Where does the middleman's fee do real damage?

- **A.** On the smallest invoices, where a fixed charge plus a percentage takes a visible share — **correct.** A fixed charge does not shrink with the payment. On an US\$18 invoice it is already two fifths of the card fee, and below about US\$12 it costs more than the percentage part outright, which is where the tail under US\$8 sits.
- **B.** On the largest invoices, because the percentage part rises with the amount being sent and the fixed part does not — The percentage does rise with the amount, but a percentage stays the same share of every payment. A large invoice absorbs it far more comfortably than a small one.
- **C.** Evenly across the run, since the same percentage is applied to every invoice the agency pays — The percentage part is even across the run. The fixed part is not, and that is the half of the fee that decides whether a small payment is worth sending at all.

- **D.** Nowhere in this run; the international bank transfer is the single charge worth discussing — The bank transfer is the most expensive single line in the exhibit, yet the agency sends hundreds of small payments rather than one large one, so the small charges add up.

2. The finance lead proposes moving the whole monthly run to the ledger-based method. What does the agency give up along with the fee?

- **A.** The confidentiality of its own banking details, which the ledger would publish next to every payment it makes — Banking details never reach the ledger. What appears there are addresses and amounts, so the privacy problem is real but it is about the permanent public record instead.
- **B.** Dispute resolution and the ability to reverse a payment, because an entry on the ledger cannot be deleted — **correct.** The safety net is the thing the fee was buying. A ledger entry is indelible, so no middleman can reverse a transfer and nobody arbitrates when a payment goes to the wrong address.
- **C.** The ability to pay translators in countries whose banking hours differ from its own — Reaching people in many countries is one of the things this method does easily, since the ledger is maintained around the clock and observes no national holidays.
- **D.** Nothing of substance, since thousands of computers verify each transaction before it joins the ledger — Verification confirms that the holder of the sending address authorised the transfer. It does not confirm that the sender meant to send that amount, or to that address.

3. A month into the new arrangement, one payment goes to a mistyped address. The finance lead asks the network to reverse it. What decides whether the money comes back?

- **A.** Whether a majority of the network votes to remove the mistaken entry from the ledger — Consensus governs which entries are added, not which are taken away. Entries once made cannot be deleted, so there is nothing here for the network to vote on.
- **B.** Whether the agency can show a payment service that the address was mistyped — The agency gave up the payment service when it gave up the fee. No middleman is left to arbitrate, which is the other half of what the saving costs.
- **C.** Whether whoever controls the receiving address chooses to send the amount back — **correct.** Nothing in the mechanism retrieves a sent amount. Recovery depends on the goodwill of whoever holds that address, which is why an address is checked before the transfer is signed.
- **D.** Whether the computers that verified the transfer can be asked to withdraw it — Those computers checked a signature, and the signature was genuine. A correctly signed transfer to the wrong address is a valid entry the network has no reason to undo.

What this case was really about. The case was a trade running in two directions. The fee buys a safety net and confidentiality, and it is charged as a fixed part plus a percentage, which is why it bites hardest on small payments. Take the middleman out and the cost goes, but so does every route back: a ledger entry is as permanent when the address was wrong as when it was right.

A payment with nobody in the middle

Cryptocurrencies are virtual currencies that no central bank issues, using cryptographic techniques to secure transactions and to generate new units of the currency. The chapter is even-handed about them: they are often described as serving mainly illicit trade, and they have various legitimate applications too.

The claim worth testing is a narrow one. The technology under the best-known cryptocurrency requires no trusted middleman, which cuts the cost of a transfer to a negligible amount and makes both very small payments and international transfers practical.

It was launched around 2008 by a developer working under a pseudonym. Payment processing is handled by thousands of computers around the world, each running the same open source software, so there is no head office, no opening hours and no national holidays.

A transfer is announced to that network, and any computer that wants to verify it may. Two words are worth having in advance: a **digital signature** proves that the holder of a sending address authorised that particular transfer, and a **block** is the batch of verified transfers added to the record in one go.

Blocks are added only when a majority of the machines holding the record accept them. The sequence below is one payment, from request to permanent record.

ORDER

The life of one payment on a public ledger

Put the six steps into the order they actually happen; each explanation names the bank function that the step replaces.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

One payment, followed from the moment it is requested to the moment it becomes permanent.

1. **The payer prepares a transfer from their own address to the payee's address** — A bank would look up an account holder in records it keeps privately. Here the sending and receiving addresses are the only identities involved, and neither is a name.
2. **The payer signs the transfer with the digital signature belonging to the sending address** — This replaces the bank's check that you are who you claim to be. The signature shows that whoever controls that address authorised this particular transfer.
3. **The transfer is broadcast to the whole decentralized peer-to-peer network** — A bank would pass an instruction privately to one other institution. The announcement instead reaches thousands of computers, none of which is in charge of the others.
4. **Any computer running the open source software verifies the signature and the transfer** — Checking is done many times over by strangers rather than once by a clearing department, which is what removes the need to trust any single participant.

5. **The verified transfer is bundled with others into a block and added to the ledger** — A bank would post a line into its own private book. This block joins one public book that every participating computer holds its own copy of.
6. **A majority of the network accepts the block and the entry becomes permanent** — There is no ledger department that can amend an entry afterwards. Changing it would need consensus from a majority of nodes, and entries once made cannot be deleted.

What indelible and public mean together

The record those computers keep is a **blockchain**: an indelible, decentralized public ledger to which transactions are added in blocks, standing as proof of every transaction ever made.

Two descriptions in common use are worth correcting here, because both hide what the design is actually doing.

Pseudonymous, not anonymous — no names appear on the ledger, only addresses, but every transaction sits there permanently for anyone to read, and an address can often be tied to a person by other means. A permanent public record cannot also be a private one.

Hashing and signatures, not secrecy — hashing links each block to the one before it, so the history cannot be quietly rewritten by anybody. Nothing is concealed, because a ledger has to be readable by everybody to be checkable by everybody.

Entries once made cannot be deleted. That is the property that makes tampering hard, and it is the same property that makes a mistake permanent: there is no middleman left to reverse a payment sent to the wrong address.

COMPARE

Three ways to settle a payment

Step through the three arrangements and read the four points under each; the third is the business use the chapter mentions rather than a currency anybody spends.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Paying through a trusted middleman — A card network or a payment service stands between the two parties.

Payer (Hands financial details to one company, not to the seller) → Middleman (Verifies, holds the details, and can reverse the payment) → Seller (Is paid, and is told only what it needs to know)

- Who verifies: one company that both sides have agreed to rely on.
- What it costs: roughly one to three percent for card processing, and sometimes several percentage points for a payment service.

- When it goes wrong: the middleman arbitrates, and the payment can be reversed.
- Who can see it: the middleman sees everything, and the record is private to the parties and their banks.

Paying on a public ledger — No company sits between the two parties; the network itself keeps the record.

Payer's address (Signs the transfer with a digital signature) → The network (Thousands of computers check the signature) → The blockchain (The block joins a record nobody can delete) → Payee's address (Receives the amount, with no account to open)

- Who verifies: any computer running the open source software, which is why no single one of them has to be trusted.
- What it costs: very little, which is what makes tiny payments and international transfers practical.
- When it goes wrong: nothing reverses it, because an entry once made cannot be deleted.
- Who can see it: everybody, permanently, which is why pseudonymous is the accurate word and anonymous is not.

A ledger shared between known partners — The business use the chapter points to: organizations in banking and technology exploring the same record-keeping for other kinds of transaction.

Known partners (Each one identified in advance, unlike an open network) → An agreed entry (Added once the partners agree that the event happened) → One shared record (A single history instead of a copy held by each firm)

- Who verifies: a defined group of organizations rather than the whole world.
- What it costs: the saving sits in reconciliation and disputes rather than in payment fees.
- When it goes wrong: the partners still need a contract, because the record proves what was entered and not whether it was fair.
- Who can see it: the partners, which suits transactions that need trust, accountability and transparency at the same time.

Working and being adopted are different questions

The mechanism works. The chapter then supplies the evidence against its own enthusiasm: in 2014 several major online companies announced that they would accept the currency, those trials were terminated, and use for ordinary purchase transactions has remained negligible.

That gap is worth carrying out of this chapter. A technology can do exactly what it claims and still not be taken up, for reasons that have little to do with whether it works.

Interest has moved instead towards the ledger itself. Organizations across banking and technology are exploring blockchain for other kinds of transaction, the chapter reports, wherever three properties are

needed at once.

Trust — the parties need a record none of them can quietly edit, which is what a ledger held by many participants provides without anybody having to vouch for anybody.

Accountability — every entry carries a signature, so who added what is settled by the record rather than argued over afterwards from two sets of private books.

Transparency — the participants read one shared history instead of reconciling separate copies, which is where most of the saving in a business setting comes from.

Remove the middleman and you remove the fee, the dispute process and the confidentiality in one motion. The question is never only whether the ledger works. It is which of those three you can afford to lose for the payment in front of you.

CHECK YOURSELF

Privacy, adoption, and what the cryptography does

Four options, one best answer; read every explanation, including the ones for the options you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A friend tells you that transactions on a public ledger are anonymous, in the way that paying with cash is anonymous. What is wrong with the description?

- **A.** Nothing is wrong; the addresses carry no names, so the comparison with cash holds — Cash leaves no record behind it. A ledger keeps a permanent public one, and that permanence is exactly what makes the two unlike each other.
- **B.** Names are attached to every entry, which makes the ledger less private than a card statement — No names appear on the ledger at all. Addresses stand in for them, which is the reason the accurate word sits between the two extremes.
- **C.** The record is permanent and public, so no name is needed for a payment to be traced — **correct.** Every transaction stays visible to anyone who looks, and an address can often be tied to a person through other information, so the privacy is partial and can be undone later.
- **D.** Entries can be read only by the computers that verified them, which is a narrower kind of privacy — The ledger is deliberately open to any computer that wants to verify it, and that openness is what allows checking without trusting anybody in particular.

2. The chapter reports that several major online companies announced in 2014 that they would accept the best-known cryptocurrency, and that those trials were later terminated. What should a manager take from that?

- **A.** That the underlying technology turned out not to work in the way it had been described — Nothing in the chapter says the mechanism failed. The ledger kept running throughout, and the account of how it works is presented as accurate.
- **B.** That a mechanism can work as designed and still fail to be taken up for ordinary purchases — **correct.** The chapter supplies this evidence against its own enthusiasm: use for ordinary purchase transactions has stayed negligible even though the technology does what it claims.
- **C.** That the companies involved were too small to give the payment method a fair trial at scale — The chapter describes them as major online companies, so scale is not the explanation, and it reports the ended trials without naming a cause.
- **D.** That the ledger cannot carry the transaction volumes a large online retailer would send it — Capacity is not the reason the chapter gives. It reports an outcome rather than a technical failure, which is why adoption and not throughput is the lesson here.

3. Which description of the cryptography behind a public ledger is accurate?

- **A.** The whole ledger is encrypted, so only participants who hold the right key can read what it contains — An encrypted ledger could not be checked by strangers, and checking by strangers is the entire mechanism that removes the need for a trusted middleman.
- **B.** Digital signatures authorise transfers, and the ledger itself is meant to be readable by anyone — **correct.** A signature proves the holder of the sending address authorised the transfer, and hashing chains each block to the one before it; openness is a design requirement rather than an oversight.
- **C.** Encryption hides the amounts, while the addresses involved are left in readable form — Amounts are visible alongside the addresses. Hiding them would stop any computer from confirming that a sender actually holds what is being spent.
- **D.** The ledger is kept in one secured location and copied out to participants on request — A single secured copy is what a bank keeps. The design here is that thousands of computers hold the record at once, which is what makes tampering hard.

Card-not-present: chargebacks, screening, and fraud signals

Two people are nervous in every online sale. The buyer is sending a card number to a seller they may know nothing about. The seller is taking one from somebody they cannot see. This section is the seller's half of that worry.

Authorization is not identification

Every card transaction is authorized by the issuer. Be exact about what that answer contains.

Authorization means the card was not reported lost or stolen. It does not mean the person placing the order is the cardholder.

In a **card-not-present** sale there is no imprint and no signature, so nothing establishes identity. One protection is time: the transaction date is the shipping date, which leaves a day or more to check. Sell a download and the window is gone.

Another is the **card security code**, the three-digit number on the back of the card and deliberately kept out of the magnetic stripe. A match implies the buyer held the card. Card company rules forbid storing it.

Signals, scores, and the cost of being wrong

A fraud-screening service turns an order into a **risk score**, and the merchant decides what that number is worth. The chapter names the variables it weighs.

Address and telephone agreement — whether shipping address, billing address and phone number match what the issuer holds.

Timing — the hour the order arrived, read against the time zone the customer appears to be in.

Transaction volume — how much is being bought, set against what this shop and this buyer normally do.

Network address — the customer's address on the network and the geographic location it resolves to.

The chapter also passes on one card network's published indicators; it names Visa. Two groups concern appearance: meaningless email strings at free providers, and addresses that disagree, reach high-risk destinations, or misspell street names.

The third is pattern: orders larger than usual, several of the same item, mostly big-ticket goods, repeated orders on one card, several cards shipping to one address, and overnight delivery, which shortens the seller's window.

None of these is a test. Each moves a score, and most describe enormous numbers of ordinary customers. The cost of refusing a real buyer — the sale, the margin, the customer — belongs beside the cost of fraud.

Below are single facts about one order. Decide which way each moves the score.

SORT

Which way does this move the score?

Place each single fact about an order under what it does to a risk score; the orders described are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Raises the risk score — moves the order toward a call or a hold, without proving anything about anybody

- Four orders today on four different cards, all delivering to one address — A single stolen card explains one order. Several cards converging on one delivery address in a short period is among the transaction patterns the chapter lists, and it is a shape ordinary households do not produce.
- Overnight delivery requested on a large order from a first-time buyer — Fast shipping shortens the window the merchant has to verify anything and moves resellable goods quickly. It is also a service that impatient legitimate buyers pay extra for every day of the week.
- Billing address in one country, shipping address in another, with a misspelled street name — The chapter names shipments to high-risk destinations and misspellings of common words or street names together. Neither settles anything: people travel, forward parcels to family, and mistype their own street.
- Six of the same lens ordered at once, more than the shop usually sells in a month — Orders larger than normal and multiple items of the same type are both listed patterns, because quantity is easier to resell. A rental company placing a genuine order looks exactly the same.
- The buyer's network address resolves to a country the shop rarely ships to — The customer's network address and its geographic location are among the variables a screening service weighs. Travellers, expatriates and anyone using a privacy service produce the same reading, so treat it as a nudge.
- The delivery address is a hotel in a city the buyer does not live in — The chapter's figure shows a shipment to a public location as one of the flags an analyst notices. Professionals working away from home have parcels sent to hotels constantly, so it raises the score and settles nothing.

- A meaningless string at a free mail provider, with nothing else unusual about the order — The published indicators name exactly this: a handle of meaningless characters at a free provider. It nudges the score up and does no more than that, because an enormous number of ordinary people pick a handle that means nothing to a stranger.

Lowers the risk score — makes the order look more like the ordinary customers this shop already has

- The card security code entered with the order matches — The code is not stored in the magnetic stripe, so a match implies somebody had the physical card in hand. That lowers the score without answering the question, because a stolen card is also a card in hand.
- The buyer has ordered four times in two years and never disputed a charge — History with your own shop is one of the few signals that is about this person rather than about the shape of this order. It still cannot rule out an account somebody else has taken over.
- Shipping address, billing address and telephone number all match the issuer's records — That match is the first variable the chapter lists for a screening service, and it genuinely lowers the score. It is also the easiest of the variables for a determined fraudster to satisfy.

Tells you nothing on its own — true of fraud and of very large numbers of honest buyers alike

- The card issuer authorized the transaction — Authorization means the card was not reported lost or stolen. It is not evidence about who is placing the order, so it belongs on neither side of a fraud score.
- The shopper spent eleven minutes reading the shipping policy before checking out — It feels reassuring, and it is not one of the variables a screening service weighs. Time on a page tells you about interest in the terms, not about who owns the card.

What a chargeback takes

When a cardholder disputes a transaction the merchant is financially responsible, and issuers typically charge it back. The chapter counts that loss carefully, because sellers count only the first item.

The transaction amount — the money for the sale is pulled back, which is the loss most sellers think of first.

The merchandise — the goods have already shipped and are rarely recovered, so the item goes with the payment.

The processing costs — the fees paid to move the money are not returned when the sale is unwound.

The chargeback fee — the bank charges for handling the dispute, and a high rate can raise fees further or close the merchant account.

Minimizing chargebacks is therefore a first-order concern. Several common causes are not crime at all: unclear policies, vague descriptions, shipping terms nobody can find and prices in an unexpected currency. Those are answered by rewriting a store, not by screening a buyer.

Two fears at one counter

Screening also works better the more the checkout collects, and every extra field sends shoppers away without buying. The shopper on the other side has fears of their own.

An online customer hands much personal information to a sometimes unknown merchant. The chapter's advice to them lists what a store must get right.

WHAT A CAREFUL BUYER CHECKS	WHY IT MATTERS
The encryption indicator, before anything sensitive is sent	Card details should not travel in the clear
The privacy policy	It says what the seller may do with your data
The refund and shipping terms	They decide what happens when the parcel is wrong
Records, email and statements	An error is caught by whoever reads them

The case below puts the apparatus on one morning of orders; the shop and its figures are invented.

MINI CASE

Five orders, one morning

Read the brief and the exhibit, then make the three decisions; the shop, the buyers, the orders and every figure below are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You run a two-person shop selling professional camera equipment online. Orders leave the next afternoon, which gives you about a day to check anything that looks wrong. Five orders are pending this morning, the screening service has flagged three of them, and the goods on the table are worth more than a week of margin. You decide which ones ship.

What the shop sells

Professional camera bodies, lenses and lighting

Average order

About US\$1,600, at roughly twelve percent margin

Loss on a chargeback

The sale, the goods, the processing costs and a fee

If the rate runs high

The bank can raise fees or close the merchant account

Verification window

Orders leave the next afternoon, so about a day to check

Repeat buyers

Close to half of revenue; a refused regular rarely returns

Exhibit A · This morning’s pending orders — Five orders waiting on one morning. The shop, the buyers, the addresses and every figure here are invented for practice.

ORDER	VALUE	BILLING VS SHIPPING	EMAIL	SHIPPING	OTHER DETAIL
A-101	US\$2,950	Differ; ships to a hotel two states away	First name and surname at a studio domain	Overnight	Returning buyer, five past orders, called ahead about a location shoot
A-102	US\$310	Match	Surname at a free provider	Standard	Third small order this month from the same regular
A-103	US\$1,240	Match	Surname and initials at a free provider	Standard	Fourth order today, each on a different card, all to this address
A-104	US\$620	Match	A six-character string at a free provider	Standard	First order, one item, residential address

ORDER	VALUE	BILLING VS SHIPPING	EMAIL	SHIPPING	OTHER DETAIL
A-105	US\$1,890	Differ; marked as a gift to another surname	First name and surname at a work domain	Two-day	Buyer has ordered each spring since two years ago

1. Which order should the shop hold this morning for a verification call before anything ships?

- **A.** A-101, because the goods go to a hotel on overnight service and the addresses differ — Every feature here has an innocent reading, and this buyer supplied one before ordering: a returning customer shooting on location. Holding it risks refusing exactly the customer the shop most wants.
- **B.** A-103, because four cards in one day are shipping to a single address — **correct.** One card can be stolen; four different cards converging on one delivery address in a day is the pattern the chapter names, and it is the only signal this morning that nothing else explains.
- **C.** A-104, because the email address is a meaningless string at a free provider — Plenty of legitimate customers use free providers and choose a handle that means nothing to a stranger. This moves a score slightly and settles nothing by itself.
- **D.** A-105, because the surname on the delivery address is not the buyer's — Gifts ship under other people's names constantly, and this buyer has a two-year history with the shop. Treated as proof, this indicator would refuse a great deal of ordinary business.

2. The issuer authorized all five of these orders this morning. What has the shop actually learned?

- **A.** That the cards were not reported lost or stolen when the request was made — **correct.** Authorization confirms that the card has not been reported lost or stolen and that the charge can be made; it says nothing whatever about who is typing the number.
- **B.** That the person placing each order is the cardholder named on the card — This is the misreading the chapter warns about most plainly. There is no imprint and no signature in a card-not-present order, so identity is never established by the authorization.
- **C.** That the shop cannot be charged back for any of these five transactions — Authorization does not settle liability. In a disputed card-not-present sale the merchant carries the loss, which is why the amount, the goods, the processing costs and the fee all land here.
- **D.** That the card security code matched and the buyer was holding the physical card — The security code is a separate check, and a match does suggest the card was in somebody's hand. A card in a thief's hand produces a matching code just as readily.

3. The shop considers refusing any order that matches one published indicator. What would that rule cost this business?

- **A.** Very little, since a card network published the indicators for exactly this purpose — The network publishes them as signals that move a score, not as tests. Applied as a rule they refuse the many ordinary customers who happen to share the same features.
- **B.** Four of the five orders here, most of them from customers who would have paid — **correct.** Mismatched addresses, a meaningless free-provider address, overnight shipping and a gift delivery flag four of these five orders, and three of those four read as ordinary customers whose refusal costs the sale, the margin and the buyer.
- **C.** Only the chargeback fees the shop would otherwise pay on disputed orders — Fees are the smallest number in the arithmetic. A refused genuine order costs the whole sale now and that customer's future orders as well.
- **D.** Nothing measurable, because refused buyers usually come back and order again later — A refused buyer is far more likely to order from a competitor and remember why. The chapter's advice is to verify a doubtful order or reject it deliberately, not to refuse blind.

What this case was really about. This was card-not-present risk compressed into one morning. Authorization proves only that a card was not reported lost or stolen, indicators move a score rather than settle a question, and every refusal carries a cost of its own. The order that looked worst had an explanation waiting; the one that mattered was a pattern across orders rather than a feature of any single one.

When a signal fires

The chapter offers two responses and no third: call the customer to verify the order, which it grants can be awkward for privacy, or reject it outright. A shop that does neither is guessing.

There is a market for help: fintech providers sell fraud detection, payment security, identity verification and regulatory compliance. A manager choosing among them weighs those against cost and ease of integration.

The simulation below follows one invented order to the change it eventually forces.

DECIDE

One order, from arrival to the fix

Work the four decisions in the order the supplier faced them, and read every outcome afterwards, including the ones you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A hypothetical supplier sells replacement parts for commercial refrigeration units online. One order arrives on a Tuesday evening and is still causing trouble two months later. Nothing here is a real company, a real buyer or a real dispute.

Decision 1. A US\$3,100 order for two compressor control units arrives at ten in the evening. Billing and shipping addresses differ, the buyer has asked for overnight delivery, and the screening service returns a middling score. Nothing can leave the building before tomorrow afternoon. What do you do first?

- Decline the order. Two published indicators are present and the amount is well above average. — Two indicators are two signals, not a finding. Both of them describe large numbers of legitimate buyers, and the refusal costs the sale, the margin and probably the customer, who orders somewhere else tonight.
- Ship it tonight to protect the delivery promise, and watch the account for a dispute. — This spends the only advantage the supplier had. The window exists precisely because the goods have not moved yet; once they have, the exposure is the amount, the merchandise, the processing costs and the fee.
- Use the shipping window: check the security code, find earlier orders, and call the buyer. — **the stronger call.** Right. The transaction date is the date the goods ship, so a physical order comes with a verification window that a download would not have had. Spend it on the security-code result, this buyer's earlier orders and the number on the order; the chapter's two responses are to verify or to reject.
- Ask the buyer to email photographs of the card and an identity document before anything ships. — That creates a file of sensitive images the supplier must now protect, and card rules already forbid keeping the security code. A call to the number on the order answers the same question and keeps nothing.

Decision 2. Nobody answers the call. An hour later the buyer emails an apology, confirms the order, and explains that the delivery address is a site office they are working out of this month. The security code matched, and two earlier orders sit under the same name. There is an hour to the shipping deadline.

- Ship the order, and note on the record what was checked and what the buyer said. — **the stronger call.** Right. The email comes from the address the order supplied, so it confirms the order rather than the cardholder; what carries this call is the matching security code and two earlier orders under the same name. Refusing here would be a false decline.
- Refuse anyway, because the addresses still differ and the shipping is still overnight. — The indicators have not changed, but what the supplier knows has. Refusing on a signal that is now explained is the reflex that loses good customers while leaving the fraud problem exactly where it was.
- Ship only once the buyer changes the delivery address to match the billing address. — Nothing reaches a site office that way, and the buyer has already given a reason the supplier was able to check. It is a rule that looks strict, costs sales, and stops nobody who can simply order elsewhere.
- Ship, and store the security code with the order so the next one clears faster. — Shipping is the right call; keeping the code is not. Card company rules prohibit merchants from storing it, and a stored code is worth more to an attacker than this order ever was.

Decision 3. Seven weeks later a chargeback arrives, and it is not fraud. The buyer says the control units were not the version shown on the product page and that they could not find how to return them inside the supplier's window. The money, the goods, the processing costs and a fee are all gone.

- Write the loss off quietly, since a dispute this old is not worth the paperwork. — Writing it off keeps the loss and throws away the information. A supplier that never records why disputes happen cannot tell description problems from card fraud, and will keep buying the wrong remedy.
- Block this buyer and refuse every future order with mismatched addresses. — The buyer here is a real customer with a complaint about the goods. Blocking them punishes the one person in the story who paid, and the address rule would refuse many more like them.
- Ask the screening service why its risk score did not catch this order. — There was nothing there for it to catch. Screening estimates whether the cardholder authorized the purchase; it has no view at all of whether the product page described what shipped.
- Answer the dispute with the order record, and log what actually caused it. — **the stronger call.** Right. Answer it with the record and the product page as it stood. The reason is the useful part: this dispute was produced by a product page and a returns window, not by a stolen card. The log is what tells you that once there are ten of them.

Decision 4. The supplier reviews a year of disputes. Roughly two thirds read like this one: descriptions, returns and shipping terms. The rest are card fraud. There is budget for one change this quarter. What gets it?

- Buy a stricter screening tier and hold every order the score puts above the middle. — This buys more of a defence the supplier already has, aimed at the smaller share of its losses, and every extra hold adds delay and false declines for customers who were going to pay.
- Rewrite the product pages, the returns window and the shipping terms on the site. — **the stronger call.** Right. The chapter is explicit that unclear store policies, product descriptions, shipping terms and currencies cause chargebacks and can be minimized through good web store design. Put them where a buyer sees them before paying: two thirds of this supplier's losses live there.
- Require a verification call before shipping any order above a set value. — Calls are a reasonable tool for a doubtful order, and the chapter recommends them there. As a blanket rule they slow every large sale, irritate good customers, and still say nothing about the descriptions.
- Stop shipping to any delivery address that does not match the billing address. — Gifts, offices, workshops and travelling buyers all fail that rule. It turns a signal into a policy, refusing a great deal of ordinary business to prevent a fraction of one third of the losses.

Two failures, two remedies

The chapter's security feature follows a ticketing company through a bad year. Attribution matters: the customer figure it reports came from the group advertising the stolen data, and the antitrust matter

was still before the courts.

A breach of the records — attackers took contact details, ticket sales and card data out of a database, which is a question about how data is stored, reached and monitored.

A takeover of the account — attackers entered customer accounts and moved tickets out, which is a question about how a person proves they are the account holder.

A shop that answers only one of those has answered half.

CHECK YOURSELF

Chargebacks, screening, and false declines

Four options, one best answer; read every explanation, including those for the options you did not choose. Any shop described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. An order is charged back to a hypothetical shop. Beyond the money for the sale, what else does the chapter say the merchant loses?

- **A.** The processing costs only, since the goods are returned to the seller — Disputed goods usually do not come back, and even when they do the sale, the fee and the processing costs are gone. Treating a chargeback as a return understates it badly.
- **B.** The tax already remitted on the sale and the cost of the shipping label — Tax and postage are real costs, but they are not the list the chapter gives, and naming them misses the fee and the merchandise that make a chargeback expensive.
- **C.** Nothing further, because the card issuer absorbs the merchandise loss — The merchant is financially responsible for a disputed transaction, which is exactly why issuers charge it back to the merchant rather than absorbing the loss themselves.
- **D.** The merchandise, the processing costs, and a chargeback fee — **correct.** The chapter lists four losses at once, and adds that a high chargeback rate can bring higher bank fees or the closure of the merchant account.

2. A hypothetical shop adds five more fields to its checkout so the screening service has more to weigh. What does the chapter say it should expect?

- **A.** Better screening with no effect on how many shoppers finish the checkout — Lengthy checkout is one of the reasons the chapter gives for cart abandonment, alongside security concerns and comparison shopping. More fields is not a free improvement.
- **B.** Better screening and more abandoned carts, to be weighed against each other — **correct.** The chapter says screening works best when the merchant collects as much data as possible, and that doing so may lead some customers to abandon their carts.

- **C.** Weaker screening, because the extra fields give a determined fraudster more to imitate — Extra fields do not weaken a score; more variables generally sharpen it. The cost of collecting them is paid in abandoned carts rather than in accuracy.
- **D.** No change either way, since screening rests on the network address alone — The network address is one variable among several, listed beside address agreement, timing and transaction volume. No screening service rests on that one alone.

3. A hypothetical seller decides to refuse every order whose billing and shipping addresses differ. What is the predictable result?

- **A.** Gift orders, office deliveries and travelling buyers are refused along with some fraud — **correct.** An address mismatch raises a score and describes a great deal of ordinary business, so each refusal costs the sale, the margin and usually the customer as well.
- **B.** Fraud attempts fall and the refused orders turn out to be almost entirely fraudulent — The rule cannot separate the two, because it never looks at anything else. It refuses the fraud that happens to match and every honest buyer who matches too.
- **C.** Chargebacks fall to almost nothing, because mismatched addresses cause most disputed transactions — The chapter attributes many chargebacks to unclear policies, product descriptions, shipping terms and currencies. An address rule does nothing whatever about those.
- **D.** Nothing measurable changes, since the screening service already weighs the whole order — A hard rule overrides the score rather than joining it. Once orders are refused before the number is even read, the service's weighing decides nothing.

Legal ground: consumers, tax, and intellectual property

A storefront anybody can reach also reaches into places whose rules you have never read. The chapter picks out three issues that follow, and this section walks each to the question it puts to a small seller.

Three issues, and one standing caution

The chapter names consumer protection, taxation of online purchases, and intellectual property as the three that matter most, especially for digital products. Non-compliance brings penalties and damages customer relationships.

One caution stands over all of it. Particulars differ by jurisdiction and they change, so what follows is the shape of each obligation. Whether a duty reaches your business is a question for somebody qualified.

Selling to somebody who cannot touch the goods

Offline a customer picks the thing up. Online that interaction is missing, so the buyer leans on what the seller says. The chapter puts transparency and accuracy at the centre of this.

Two duties come up in most places, in the chapter's account.

A clear return and refund policy — written so an ordinary reader understands it, and displayed prominently rather than buried three clicks down.

No deceptive practice — false advertising and misleading claims about a product's quality, benefits or price are commonly prohibited, whatever the seller intended.

Compliance keeps you out of trouble, but the chapter's reason for going further is commercial. Its example is a retailer whose generous returns cost a great deal in reverse logistics and raise revenue anyway.

Then the borderless part. The chapter reports an attempt to harmonize these rules across the European Union, while in the United States fraudulent practices sit with a federal trade regulator and other matters sit at state level.

Because many of these rules apply based on where the **customer** is, not where the seller sits, many sellers write one policy to the strictest set they expect to meet.

MINI CASE**Three new places, one small studio**

Read the brief and the exhibit, then take the three decisions in order; the studio, the regions and every condition in the exhibit are invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Two people run a studio that sells sheet music. The same arrangement goes out either as a printed score in an envelope or as a download, and the buyer chooses which. Orders have just started arriving from three regions the studio has never sold into before, and neither partner knows what selling there commits them to. Nothing has gone wrong yet.

What it sells

One catalogue of arrangements, as printed scores or as downloads

Who runs it

Two people, with no lawyer and no accountant on retainer

Home base

One region, where it has sold and collected tax for years

What is new

Orders from three regions where it has no presence at all

What each region says

Past our trigger level you collect our tax, presence or not

What is at stake

Penalties and refunds a studio this size cannot absorb

Exhibit A • the three new regions — Invented conditions, written for practice. Each region here collects from any seller past its trigger level, with or without a presence there. Nothing describes any real place's law, and none of these figures is reported data.

REGION	RETURN WINDOW BUYERS THERE EXPECT	SALES PAST THAT REGION'S TRIG
Region One	14 days, no reason needed	No, well under it
Region Two	30 days, no reason needed	Yes, crossed in March
Region Three	14 days for goods, nothing stated for files	Yes, crossed in June

1. Sales into which regions can create a duty to collect tax this year, although the studio has no presence in any of the three?

- **A.** Regions Two and Three, where the exhibit shows the trigger level passed — **correct.** Both of those rows carry a yes in the trigger column, and passing that level is the condition each region says creates the duty, whatever the studio does or does not own there.
- **B.** Region One alone, where the exhibit shows the shortest return window — Region One is the row sitting under its trigger, so it is the one place the duty has not attached. A return window is what buyers expect of a policy, and it settles nothing about tax.
- **C.** Region Three alone, where the exhibit shows the order arriving as a download — Format changes how the order is delivered, not whether a threshold was passed. Region Three does qualify, but it qualifies on its trigger row, and Region Two qualifies beside it on the same column.
- **D.** No region at all, since no office or warehouse of the studio sits in any of them — Owning a building somewhere is the older test, and the exhibit is deliberately built to defeat it. Two of these regions record a threshold crossed, which is enough on its own terms.

2. The Region Three order in the exhibit is a download rather than a parcel. What does that change?

- **A.** The tax question disappears, because nothing physically crosses a border — The chapter says close to the opposite about digital products: their movement is difficult to track, so the tax revenue is easily lost rather than never owed in the first place.
- **B.** Nothing at all changes, since a file and a parcel are handled identically — Format changes real things here. The envelope needs a carrier and a customs declaration, while the download needs neither and arrives before anybody could inspect anything.
- **C.** Delivery stops being a shipping problem; the tax question gets harder — **correct.** The file removes the shipping step and the border formalities that come with it, and leaves the harder half standing: an obligation can still attach where the buyer is, and nobody can watch a download cross anything.
- **D.** The buyer loses any expectation of a refund, because a file cannot be posted back — Refund expectations do not vanish with the packaging. The chapter treats clear return and refund terms as a general duty, and many sellers offer a window on files as well.

3. The partners want to act this week. What should the studio actually do next?

- **A.** Copy the refund and tax wording from a larger seller into their own storefront — A larger seller's wording was written for its own footprint and its own advice. Copying it transfers none of that reasoning while still binding the studio to whatever it now promises.
- **B.** Ask somebody qualified in each region what applies, and write the answer down — **correct.** The particulars differ by jurisdiction and they change, so the reliable move is to find out from somebody qualified in that place rather than guess, and to record what was decided so the next order is not a fresh argument.
- **C.** Wait for the first complaint, then deal with whichever region it comes from — A complaint is a slow and expensive detector. By the time one arrives the studio has been trading under terms it never checked, and the repair now comes with an unhappy customer attached.

- **D.** Stop selling into the two regions whose conditions look complicated — Withdrawing is a legitimate business choice and sometimes the right one, but it answers a different question. It also gives up revenue the studio may well keep once it knows what the region asks.

What this case was really about. One column did most of the work. The trigger, not the address, decided where tax could attach, and the studio owns nothing in either region that crossed it. The download changed the delivery and made the tax harder to see without making it go away, and the widest return window in the exhibit is the one a single policy has to clear. The threshold rule has a name in the chapter, and the next stretch of reading supplies it.

Where a tax can attach

The trigger column the studio just read has a name behind it. Online sales were once treated like mail order: tax was collected only from customers in a state where the business had substantial presence, so firms chose home bases offering effectively tax-free shopping.

The chapter reports that a 2018 Supreme Court ruling changed that shape. States may collect from sellers with no physical location once certain sales thresholds are reached, known as economic nexus. Tracking volume and collecting the tax becomes the seller's work.

Crossing a border makes it harder again, and three patterns are worth holding.

A duty can follow the buyer home — a customer ordering from abroad may owe use tax in their own state on what they bought.

Arrival can trigger a charge — a customer outside the country pays no sales tax to that country, yet may owe tax or an import duty when the shipment lands.

Digital products slip through — the movement of a download is difficult to track, and the chapter notes that the tax revenue is easily lost.

Owning content, and not borrowing it by accident

Ownership of content and trademarks is, in the chapter's phrase, a virtual minefield. It runs in two directions, and sellers usually think about only one.

Do not infringe — posting product photographs without permission, or putting somebody else's trademarked name inside your own domain, are the chapter's examples.

Settle ownership in advance — when a third party builds content for your site, agree who will own the copyright before the work starts.

Protect your own — copyright notices and the like, because property nobody manages can mean lost revenue, brand dilution or legal action.

The chapter closes with a housekeeping instruction that is easy to skip: state your jurisdiction clearly, and comply with it.

SORT

Which of the three issues is this?

Drop each situation into the issue it raises first; every seller and situation described here is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Consumer protection — what the buyer is told, what they are shown, and what they may undo

- A returns page most readers will not find before they buy — The chapter asks for policies that are clear, understandable and prominently displayed. A policy that exists but hides fails the display half, which is the half a buyer actually relies on.
- A description promising a benefit the item does not deliver — Misleading claims about quality, benefits or price are the deceptive practices the chapter says are commonly prohibited. Being sincerely mistaken does not move it out of this bucket.
- A price shown as a reduction from an amount nobody was charged — This is a misleading claim about price rather than a dispute over ownership or a duty owed to a treasury, which is what keeps it here instead of in either neighbour.

Tax — what is owed on a sale, to whom, and what happens once a line is crossed

- Keeping a running total of what has been sold into each region — Economic nexus turns volume into a duty, so the counting exists because a threshold might be crossed. The chapter is explicit that this tracking burden lands on the seller.
- A parcel held on arrival until the buyer settles a charge at home — The chapter describes a buyer abroad paying no sales tax to the seller's country while still owing tax or an import duty at home. It looks like a delivery problem and is not one.
- Choosing where to put a warehouse partly for what it means at checkout — Substantial presence used to decide who had to collect, so the location of a building was a tax decision dressed as a logistics one. The chapter reports that a 2018 ruling narrowed how well that works.
- A download whose movement between countries nobody can trace — The chapter raises this as a taxation difficulty: the product moves invisibly and the revenue is easily lost. The same file being easy to copy is a separate issue, and it belongs one bucket over.

Intellectual property — who owns the work, the mark or the file, and what the owner may control

- Agreeing in writing, before work starts, who will own a freelancer's pages — The chapter tells sellers to settle copyright with third parties who develop content for the site. Left unsettled it becomes a dispute about ownership, not about what a customer was told.
- Putting a copyright notice on the studio's own catalogue photographs — Protecting your own property is the half sellers forget. The chapter warns that failing to manage it proactively can mean lost revenue, brand dilution or legal action.
- A bought album that plays on a phone but not on the buyer's new laptop — Device limits are digital rights management, a rights holder controlling its own media. Critics argue it is really a consumer rights matter, which is exactly why it sits so close to the line.

Locks on a file

Digital media copy perfectly and cost nothing to pass on, a problem publishers took to a technical answer. **Digital rights management** is a technological solution letting publishers control their digital media to discourage, limit or prevent illegal copying.

The restrictions cover which devices will play the media, how many devices it plays on, and how many times it can be played.

Content sold without them is often watermarked instead. A **digital watermark** is an electronic version of the mark on paper currency, and the chapter's example is a purchased music file carrying the buyer's email address, so an illegal copy traces to its buyer.

MATCH

Which issue does this situation raise?

Pair each situation with the issue it raises first, then read why the neighbouring issue is not the answer; every business described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The refund terms sit three clicks deep behind a collapsed menu at the foot of the page

Consumer protection The chapter asks for return and refund terms that are clear and prominently displayed, so burying them is the failure even when the wording itself is generous. Nothing here turns on who owns the page, which would be the intellectual property question instead.

Sales into a region where the seller has no office or warehouse pass that region's trigger level

Tax This is economic nexus in the chapter's account: crossing a sales threshold can create a duty to collect with no physical presence at all. Its neighbour, consumer protection, follows the buyer's location rather than a volume of sales.

A supplier's product photograph is lifted onto the storefront without asking

Copyright Copyright covers the photograph as a work somebody made, and posting product images without permission is the chapter's own example. The neighbouring issue is trademark,

which covers a name or mark identifying a business rather than the content itself.

A new store domain is registered with another firm's brand name inside it

Trademark Trademark protects the identifier rather than the content, and the chapter names a trademarked name in a domain as its example. Copying that same firm's photographs would be the copyright half of the identical mistake.

A music file sold with no restrictions at all carries the buyer's email address inside it

Digital watermark The chapter describes content sold without restrictions being marked this way so that an illegal copy traces back to the person who bought it. Digital rights management is the neighbour, and the difference is the whole point: restrictions try to prevent a copy, while a mark only says where an existing copy came from.

The argument, kept even

Both sides are worth stating plainly, and the chapter settles nothing.

- The industry says the restrictions minimize sales losses from unauthorized duplication.
- They add that it infringes existing consumer rights and stifles innovation.
- That inconvenience, critics argue, can breed the piracy it was built to stop.
- Critics rename it digital restriction management and call enforcement arbitrary.
- Activation and device limits inconvenience honest buyers, as on a new computer.
- And sellers who dropped the restrictions entirely have been very successful.

Legal ground is not a wall you hit once. It is a set of questions that attach to where your customer is, what you sold them, and whose work is on the page — and most answers begin with finding out.

CHECK YOURSELF

Location, thresholds, and a fight nobody won

Four options, one best answer; read every explanation, including the ones for options you did not choose. Any seller described is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Why do many online sellers write one policy to the strictest standard they expect to meet, rather than a different policy for each place they ship to?

- **A.** Because a federal regulator requires a single national policy from every online seller — The chapter describes a federal trade regulator handling fraudulent business practices while other matters are handled at state level, which is a division of authority rather than a demand for one national policy.

- **B.** Because many of these rules follow where the buyer is, not where the seller is — **correct.** Many consumer protection rules apply based on the location of the consumer, whether or not the business has a presence there, so a single policy pitched at the strictest expected standard is cheaper to run than one policy per region.
- **C.** Because a strict policy lowers the cost of returns and reduces refund requests — A generous policy usually raises the cost of returns rather than lowering it. The chapter's own example accepts high reverse logistics costs in exchange for revenue over a longer horizon.
- **D.** Because penalties can come from the strictest region and from nowhere else — Exposure is not limited to one strict region. The difficulty the chapter describes is precisely that rules can attach through the buyer's location, so several sets may apply at once.

2. A seller has no office, staff or warehouse in a region, and this year its sales into that region pass the level that region uses as a trigger. What does the chapter report changed in 2018?

- **A.** Sellers became liable for tax in every region their website can be viewed from — Being reachable is not the test. The chapter describes thresholds measured on sales into a place, which is a quantity somebody can count rather than the mere availability of a website.
- **B.** Physical presence stopped being the only thing that can create a duty to collect — **correct.** The chapter reports a 2018 Supreme Court ruling under which states may collect from online sellers with no physical location once certain sales thresholds are reached, and calls that economic nexus. Particulars differ by jurisdiction and change, so a seller confirms them locally.
- **C.** Online sales were reclassified as mail order, which set the threshold rule aside — Mail order is the older treatment the chapter describes, under which collection followed substantial presence. The ruling moved away from that arrangement rather than restoring it.
- **D.** The duty to collect moved from the seller to the buyer once a threshold was crossed — Buyers can owe use tax in some of the situations the chapter describes, but this change runs the other way: it places the tracking and the collecting on the seller.

3. How does the chapter leave the argument about digital rights management?

- **A.** Settled for the industry, since the restrictions demonstrably cut illegal copying — The chapter records the industry's claim that these restrictions minimize sales losses from unauthorized duplication, but it reports that as an argument rather than as a demonstrated result.
- **B.** Settled for the critics, since sellers without restrictions have done well — Critics do point to successful sellers of unrestricted content, and the chapter includes that. It is one of several claims on that side, not a verdict the chapter hands down.
- **C.** Open, with a claim on each side: fewer illegal copies against real inconvenience — **correct.** The chapter sets the industry's case about unauthorized duplication beside the critics' case about arbitrary enforcement, consumer rights, stifled innovation and device limits that inconvenience honest buyers, and it chooses between them nowhere.

- **D.** Beside the point, since watermarking already traces a copy back to its buyer —
Watermarking is what the chapter describes for content sold without those restrictions, so it is the alternative under discussion rather than a reason the disagreement stops mattering.

The ecosystem behind a storefront, and the store on the street

Putting something on sale online is not one decision but four: where the store lives, where customers come from, how the money moves, and who packs the box. This section takes them in order, then walks into a physical shop the same technologies reached.

Application supplement. This section assembles the chapter's own Technology Today and Industry Analysis features, which sit outside the two learning objectives the supplied chapter carries. It is practice with chapter material, not a further objective.

Four layers behind a storefront

Competition and customer expectations have both risen, and almost nobody answers them alone. The chapter describes an ecosystem of key players supplying the infrastructure for online sales, payments and logistics — four layers, in the order a seller meets them.

Storefront platform — software that creates and manages the store itself, ranging from intuitive tools aimed at small and medium businesses to customizable stores that slot into a website a company already runs.

Marketplace — somebody else's shopfront, which lends a seller an existing customer base and technology and connects a new listing with millions of shoppers on the day it goes up.

Payment provider — the layer that takes the money: a widely trusted first mover, an integration a developer can shape, card readers for a small counter, or one-touch checkout on a phone.

Fulfillment provider — warehousing, stock, shipping and returns handled by an outside firm, offering speed and scale without the upfront investment of a building and the staff to run it.

Notice what each layer leaves standing. A platform builds the store but brings nobody to it, which is hardest for a fledgling business. A marketplace fills that gap, and rents it.

EXPLORE

The four layers, one at a time

Open each layer and read all four facets before moving on; the sellers described in them are hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Storefront platform · the store itself

- **What it gives you:** Pages, a catalogue, a cart and a checkout you did not have to build, on tools ranging from intuitive ones for small and medium businesses to customizable stores that fit an existing website.
- **What it costs you:** A running fee and a cut of each sale, plus the work of operating a shop that is genuinely yours to operate.
- **What it does not solve:** Nobody arrives because the store exists. Attracting customers to the site remains entirely your problem, and it is hardest for a business nobody knows yet.
- **When to add it:** First, as soon as you want somewhere of your own to send people, and a customer list you keep rather than borrow.

Marketplace · somebody else's crowd

- **What it gives you:** An existing customer base and technology, connecting a listing with millions of shoppers on the day it goes up rather than in the year it takes to build an audience.
- **What it costs you:** A larger share of every sale, and a buyer who belongs to the marketplace rather than to you.
- **What it does not solve:** It does not make you findable among everything else listed there, and it does not hand you a relationship you can sell into again next season.
- **When to add it:** When the store is ready but the traffic is not, or when a new product needs testing in front of demand that already exists.

Payment provider · the money moving

- **What it gives you:** Checkout that shoppers already trust: a widely accepted first mover, integrations shaped to fit a site, card readers for a small counter, or one-touch checkout on a phone.
- **What it costs you:** A percentage of each transaction, and a dependence on somebody else's rules about disputes, holds and account standing.
- **What it does not solve:** It does not judge whether a transaction is genuine, and it does not remove the merchant's exposure when a cardholder later disputes the sale.
- **When to add it:** Before the first sale. Of the four layers this is the one no online seller gets to postpone.

Third-party fulfillment · warehousing, packing, returns

- **What it gives you:** Warehousing, stock handling, shipping and returns run by a firm that already has the buildings, the vans and the people standing by.
- **What it costs you:** A fee for each unit stored and shipped, and much less direct contact with how the parcel actually arrives.
- **What it does not solve:** It does not fix a product people do not want, and it does not take your name off a delivery that turns up late or broken.
- **When to add it:** When packing has outgrown the space and the hours available — the chapter's image of a successful seller outgrowing the garage.

Selling also requires having something to sell. Sourcing marketplaces, weighted heavily towards business-to-business trade, exist for that half of the problem: access to manufacturers, suppliers and markets abroad.

Nobody buys all four layers at once, and the order matters. The case below puts that sequence to a hypothetical maker of hand-thrown tableware who sells at craft fairs; the business and its figures are invented for practice.

MINI CASE

A potter decides how to go online

Read the brief and the exhibit, then work the three decisions; the business, the routes and every figure below are hypothetical and invented for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A potter makes hand-thrown tableware alone and sells it at craft fairs six weekends a year. Demand at the last two fairs outran what she brought. She wants to sell between fairs, has no customer list beyond a notebook of names, and packs every box herself at the kitchen table. She has to choose a route online before the next making season starts.

Made each month

About 90 pieces, by one person

Sold at fairs

Around 60 pieces a month across the season

Average price

US\$38 a piece

Packing

About 12 minutes a piece, boxed and labelled

Breakage in transit

Roughly 1 shipped piece in 20

Exhibit A · four routes online — Four routes the potter is weighing, side by side. Every figure here is invented for practice rather than reported from any real seller or platform.

ROUTE	CUSTOMERS REACHED	FEE TAKEN PER SALE	WHO OWNS THE CUSTOMER BASE
Her own storefront on a platform	Only the people she brings herself	About 3% plus a monthly fee	She does, with the email list
A listing on a large marketplace	Millions already shopping there	About 15% of each sale	The marketplace does
Both together	Both, and fair regulars land on the store	Around 9% blended across the mix	Split, and she keeps the store side
Craft fairs only	Whoever walks past the stall	Stall rent, near 8% of takings	She does, but face to face only

1. She wants more sales than the fairs produce, without renting out her whole customer base permanently. Reading the exhibit, which route does it actually favour?

- **A.** Craft fairs only, because no platform stands between her and the buyer — Nothing in that row is untrue: no platform sits between her and the buyer, and she keeps every name she meets. The customers-reached cell is what settles it, since only whoever walks past the stall six weekends a year can buy, which is the shortfall she is trying to fix.
- **B.** The marketplace listing alone, because that row reaches the most people — Reach without ownership is the marketplace bargain, and taken alone it hands the buyer to somebody else at the highest fee in the exhibit. It answers her sales problem and worsens the one she named about her customer base.
- **C.** Her own storefront alone, because that row keeps the customer relationship — The storefront row keeps the relationship, which is what she asked to protect, but its customers-reached cell says only the people she brings. That is the layer's known gap: a platform builds a store and brings nobody to it.
- **D.** Both together, because one row supplies reach and the other keeps the list — **correct.** The marketplace supplies the customer base the store cannot, while the store keeps a list she owns and converts the fair regulars, and the blended fee sits between the two rows rather than at either extreme.

2. Within a week of listing, marketplace orders arrive from people who have never heard of her. In the chapter's terms, what has she rented, and what does the marketplace keep?

- **A.** It lends warehouse space, and keeps the right to set her prices — Storage is a fulfillment service, which a marketplace may sell separately but is not what a listing itself provides. Pricing also stays with the seller here, so both halves of this option describe something other than the bargain she just made.
- **B.** It lends a payment provider, and keeps the tax duty on each sale — A marketplace does route the payment, but the payment layer is a distinct part of the ecosystem, and a duty to collect tax attaches to the transaction and the jurisdiction rather than being handed over with a listing.
- **C.** It lends its customer base and technology, and keeps the buyer — **correct.** The chapter describes marketplaces as lending an existing customer base and technology; the seller gets instant connection to shoppers, and the buying relationship stays with the marketplace rather than transferring to her.
- **D.** It lends her reputation with shoppers, and keeps a share of stock — Reputation is a real benefit of listing somewhere trusted, but it is a by-product rather than the thing rented, and the marketplace takes a share of the sale price, not a share of the pieces she has made.

3. In the months between fairs, online orders settle at roughly 85 a month. At what moment does paying a third party to warehouse, pack and ship stop being an added cost and become the only way to grow?

- **A.** When the packing hours have started eating the hours she makes work in — **correct.** At twelve minutes a piece, 85 orders is about seventeen hours of packing a month, two full working days taken from the only person who can make the product. Fulfillment stops competing with her margin and starts competing with her output, which is the chapter's point about sellers outgrowing the garage.
- **B.** When a provider quotes a fee below what her own boxes and tape cost — Comparing the fee against packaging materials leaves out the labour, which is the expensive part here and the part that cannot be bought back. On that comparison alone a third party will almost never look worth paying for.
- **C.** When she passes the monthly order count a provider treats as a minimum — A provider's minimum tells her when she is allowed to buy the service, not when she needs it. Eligibility is a supplier's rule; the decision belongs to what her own hours are being spent on.
- **D.** When breakages rise, since a warehouse packs more carefully than she can — Fewer breakages would be a genuine benefit and worth pricing, but it is an improvement to a service she is already delivering. It does not change what limits her growth, which is the time one person has.

What this case was really about. The exhibit looks like a fee comparison and is really a comparison of reach, ownership and time. Platforms build a store and bring nobody; marketplaces lend a customer base and keep the buyer; fulfillment converts hours back into making. That is the chapter's ecosystem read from the seller's side, and the same three columns apply well beyond tableware.

The store on the street

Now the shop you can walk into. Decades ago, large chains installed computerized point-of-sale inventory systems: a bar code scan captures the sale and deducts the item from stock, so the retailer sees in real time what to reorder.

Two things follow from that scan. Checkout gets quicker and shelves run empty less often, which customers feel as satisfaction rather than as a system. Every later change moved the scan.

- Self-checkout, saving time and labour cost
- Scanning into the cart, paying at the counter
- Tags read by radio at a distance
- A store that watches the shelf instead

A Swiss grocer built the second: customers scan into the cart, then pay at the counter. The third totals a full cart in about a second with no scanning, but only where an item is worth the cost of a tag.

The fourth skips tags entirely. Radar and computer vision detect what a customer takes off the shelf; the customer taps a card to enter, then leaves.

Payment changed on the shop floor too, in three directions.

Multi-factor authorization — more than one proof of who is paying, drawn from a card, phone, code, signature, fingerprint or face.

Chip and touchless cards — the card carries a small processor rather than only a magnetic stripe, and it no longer has to be handed over or swiped.

Near-field communication — the customer's phone and the retailer's terminal talk across a few centimetres, and the amount goes to the payment method already stored on the phone.

Three journeys are worth setting side by side: a customer through such a store, the same customer online, and the case where the two meet. Each has a moment where the retailer learns, and one where it loses the sale.

COMPARE**Three journeys past the same shelf**

Step through each journey and read the points beneath it, watching for where the retailer learns something and where it loses the sale; the sellers pictured are hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A customer through a store with point-of-sale technology — The scan is the moment the system learns that anything happened at all.

Walks in (Nothing recorded yet; the shelf is doing the selling) → Picks up the item (Stock still reads whatever the last scan left) → Scans at the counter (Sale captured, item deducted, reorder signal raised) → Taps to pay (Terminal and phone settle against a stored method)

- The retailer learns at the scan and not before it, so everything earlier in this journey is invisible to the inventory system.
- It can intervene at the shelf and at the counter, which is why staff, offers and impulse goods are placed at exactly those two spots.
- It loses the sale before the journey starts, when the item is not on the shelf. Reducing that is what real-time tracking was installed to do.
- Self-checkout and scan-into-the-cart move the scan earlier or onto the customer; a checkout-free store moves the learning moment to the shelf itself.

The same customer online — Every step is recorded, and every step is also a place to leave.

Arrives from a search (The seller pays for that arrival one way or another) → Fills a cart (Recorded, and abandoned more often than not) → Checkout (The payment provider takes the sensitive part) → Fulfillment (Somebody else may pack the box being judged)

- The retailer learns at every step, including the steps where nothing was bought, which is the sharpest difference from the aisle.
- It can intervene during checkout, which is why the number of fields asked for there is a commercial decision and not a technical one.
- It loses the sale in the cart: the chapter reports that more than half of online shopping carts are abandoned before payment.
- The parcel is the only physical contact in the whole journey, and it is the layer most often handed to a third party.

Showrooming — The store pays for the moment that convinces the customer to buy elsewhere.

Examines the goods (In person, in the aisle, at the store's expense) → Checks prices on a phone (Comparing against sellers with no storefront to fund) → Orders elsewhere (The cost of the building is never recouped)

- The retailer learns nothing at all here unless a member of staff asks, which is why the human element keeps reappearing in this argument.
- It can intervene only while the customer is still standing there, so the window is minutes long rather than days.
- It loses the sale at the phone, having already paid for the display, the space and the person who answered the question.
- Two large chains named in the chapter answer at the middle box instead: invite the comparison, match the price while the customer is still in the aisle, and sell something else alongside.

What the shop still has to answer

That third journey has a name. In showrooming, a customer examines a product in person, then leaves to order it online for less. Storefronts cost billions to build and maintain, an online seller carries no such cost, and the store never recoups it.

Two large chains named in the chapter encouraged the comparison rather than fighting it: browse the shelves, check prices online, and they will match the price and add personal service, keeping the customer and selling something else alongside.

Other in-store technology pushes the same way. Fitting rooms use augmented reality to show how an item would look or to suggest something to go with it, and small radio beacons hand a shopper an offer while the goods are in front of them.

The chapter closes with three questions rather than answers. How can technology help a physical retailer compete with an online seller? How should it answer the criticism that tags track purchasing habits too closely? And how does a store avoid feeling sterile as people are replaced?

Every technology here moves work somewhere: to the customer, to a supplier, or to a sensor. The decision is not whether the work moves, but what the store puts in the place it left.

CHECK YOURSELF

The chapter's three closing questions

Four options, one best answer; read every explanation, including those for options you did not choose, and treat any store described as hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A hypothetical chain finds customers examining goods in its aisles and then ordering the same items online for less. Which response matches the one the chapter describes as working?

- **A.** Match the online price at the counter and add service a screen cannot give — **correct.** The chapter names retailers that embraced showrooming by inviting customers to browse and compare, then holding the sale with price matching and superior personal service, and selling additional products alongside it.
- **B.** Ask floor staff to discourage customers from comparing prices on their phones — Policing the phone treats the customer as the problem and removes the one advantage the store has left, which is that the customer is physically present and can be helped in the aisle.
- **C.** Reduce the range to items that online sellers tend not to stock at all — Narrowing the range is a real strategy in some categories, but it concedes the comparison rather than answering it, and it shrinks the reason a customer came into the building.
- **D.** Close the quieter branches and move that part of the budget into the website — Shifting budget online abandons the storefront the chapter says has already been paid for, and it puts the chain into direct price competition with sellers that never had that cost.

2. Privacy advocates object that tags read by radio let a retailer track purchasing habits closely. Which response addresses the objection rather than restating it?

- **A.** Note that the readings never leave the retailer's own systems and databases — Keeping data in-house is confidentiality, not consent. The objection is about a retailer building a record of somebody's habits, and that record is built whether or not anyone outside the company sees it.
- **B.** Say what is collected, offer a genuine opt-out, and kill the tag at the till — **correct.** Disclosure, a real choice and deactivating the tag when the customer leaves the store together answer what is collected, whether the shopper agreed, and how long the tracking can continue.
- **C.** Store the readings under a customer number instead of the shopper's name — A number rather than a name makes the record pseudonymous, not anonymous: purchases still accumulate against one shopper, and a loyalty account or a card ties that number back to a person.
- **D.** Explain that the tags exist to keep the shelves stocked, not to watch anyone — Stating a purpose is not a control. An intention can be sincere and the data still be retained, combined with other records, or used later for something the shopper was never told about.

3. A hypothetical grocer replaces staffed counters with self-checkout and cuts floor staff. Sales hold, but shoppers call the place sterile. What does the chapter's third question ask the grocer to weigh?

- **A.** Whether the machines are quick enough to justify what the change cost — Speed was the original case for self-checkout and the machines here are presumably delivering it. The complaint is not about how long the queue takes, so a faster machine leaves it entirely unaddressed.
- **B.** Whether competing grocers have removed a similar number of counter staff — What rivals have done sets a benchmark for cost, not for the experience in this store. Matching another grocer's staffing says nothing about which of these shoppers wanted help and did not get it.

- **C.** Which moments customers actually want a person for, and staffing those — **correct.** The chapter asks how a store avoids becoming too sterile as the human element is replaced, which is a question about placing people where they are wanted rather than about removing or restoring them wholesale.
- **D.** Whether the savings are large enough to fund a further round of automation — Funding more automation answers a finance question with the same move that produced the complaint, and it treats the savings as the goal rather than as one input into the decision.

Turning this chapter into an analysis

An observation is not a diagnosis. A firm can count abandoned baskets, disputed payments and marketplace commission and still hold nothing anybody can act on, because a figure never says which pressure produced it. This supplement practises the move from counted evidence to one recommendation.

What this section is. It is an application supplement rather than a textbook objective: it teaches no new framework, and it assumes you have already met the five competitive forces and the value chain. Everything here runs on a hypothetical firm, and it solves no graded case for you.

The firm you are advising

The firm below is hypothetical and every figure attached to it is invented. It is a regional supplier of laboratory and workshop consumables — gloves, glassware, abrasives, safety kit — that has sold on account against a purchase order for thirty years, to schools, clinics and small workshops.

Eighteen months ago it added a public website that takes cards, and a listing on a large marketplace. Take the case first. The explanation after it reads as the debrief of what you have just tried.

MINI CASE

Two new channels, one initiative

Read the brief and the exhibit, then take the three decisions in order; the supplier and every figure attached to it are hypothetical, invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A regional supplier of laboratory and workshop consumables has sold gloves, glassware, abrasives and safety kit for thirty years, almost all of it on account against a purchase order. Eighteen months ago it added a public website that takes cards and a listing on a large marketplace. Both channels sell, and both carry costs, risks and duties the account book never had. The owners can fund one initiative this year. The firm is hypothetical and its figures are invented.

The firm

A regional supplier of laboratory and workshop consumables

Sells to

Schools, clinics, repair shops and small manufacturers

Sells through

Account orders, a trade counter, its own website, a marketplace listing

What changed

The two online channels are eighteen months old

The ask

One initiative, with what it changes and what it costs

Exhibit A • The first full year on the new channels — Five things the owners can count, with the figure each one produced. Every figure here is invented for practice.

WHAT THE OWNERS COUNTED	THIS YEAR	WHAT IGNORING IT COSTS
Cardholder disputes on website orders	9 of 552 orders, against 4 of 1,305 on the marketplace	Fees, lost goods, and a bank that reprices the account
Website baskets abandoned before payment	68 in every 100 started	Traffic the supplier has already paid for produces no order
What the marketplace keeps from each order it carries	15 percent, against under 3 percent on the firm's own channels	Growth arrives in the channel that pays back least
Places goods are now delivered to and tax may be owed	14 jurisdictions, against 2 before the website opened	Compliance work grows quietly with every new place
Revenue still placed on account or over the trade counter	72 percent of the year's total	A year spent on the 28 percent that is newest

1. The maker whose gloves the supplier has distributed for years opens its own website and sells the same cases direct to the same schools. What should the analysis call this?

- **A.** Rivalry among distributors, since a competing wholesaler undercut the firm — Rivalry is pressure from firms selling the same goods into the same market, and nothing here says another distributor took the order rather than the maker itself.
- **B.** Disintermediation: a supplier now reaches the end customer directly — **correct**. The maker has cut out the middle of the chain and now reaches the end customer itself, which is disintermediation, and it raises the threat of new entrants against this firm.
- **C.** A substitute product, since the schools are buying a different kind of glove — A substitute meets the same need with a different product, such as a washable coat instead of a disposable one. The same case of gloves by another route is not a substitute.

- **D.** Buyer power, since the schools have negotiated a lower price this year — Buyer power would show as price pressure from the schools themselves. Here the price they face moved because a new seller appeared, not because they bargained for it.

2. Sixty-eight of every hundred website baskets are abandoned before payment. Where in the value chain would a fix for that actually land?

- **A.** Inbound logistics, because stock has to arrive before anybody can order it — Stock availability is a real cause of lost sales, but it stops the order before a basket exists rather than at the moment payment is asked for and the visitor walks away.
- **B.** Procurement, because better buying terms would let the firm post lower prices — Procurement decides what goods cost the firm. It can move a list price over months, and it does nothing for a visitor who has already chosen an item and then stopped.
- **C.** Marketing and sales with outbound logistics, where checkout and delivery sit — **correct.** The basket is abandoned at the checkout, so the fix lives where the offer, the payment options and the delivery promise are made: marketing and sales, and outbound logistics.
- **D.** Human resource management, because counter staff need training in online service — Training matters at the trade counter, and it is a genuine support activity, but nobody at the counter ever meets the website visitor who leaves without paying.

3. The owners can fund one initiative this year. Which should the analysis recommend, given the evidence in the exhibit?

- **A.** Rebuild the website, join a second marketplace and add a loyalty scheme together — Three initiatives at once is the commonest weak recommendation. It hides which pressure is being answered, it resists costing, and afterwards no result can be attributed to any of the three.
- **B.** Match the cheapest marketplace price on every consumable the firm lists online — Matching a marketplace price on light, postable goods means competing on ground the entrant chose, against a cost base built for a depot, a counter and a delivery van.
- **C.** Buy fraud screening and set it to refuse anything the risk score marks unusual — Screening set to refuse anything unusual trades one cost for another. Nine disputes in a year is a small number, and a hard threshold declines paying customers whose orders merely look odd.
- **D.** Move account reordering onto the firm's own site, at agreed prices, on invoice — **correct.** It defends the seventy-two percent the firm still holds against a maker selling direct, and an invoiced reorder carries no marketplace commission and no card dispute.

What this case was really about. Every decision here made the same move: refuse the symptom, name the pressure, then place it in the value chain before recommending anything. The strongest recommendation was not the most technical one. It was the one aimed at the pressure the evidence actually showed, on ground the firm already holds, and it can be stated with a cost attached.

What this chapter supplies to the frameworks

Each decision asked for the same two moves: name the pressure an observation belongs to, then say where in the firm a fix would land. A symptom is not a force. Six ideas from this chapter make that translation for you.

Disintermediation — when a maker sells direct to the end customer an intermediary disappears, which changes the threat of new entrants and the bargaining power of buyers rather than merely adding a channel.

Selling through a marketplace — the marketplace lends you its customers and keeps the relationship, which makes it a supplier whose power over you grows with every order it carries.

Cart abandonment and checkout friction — the point where a visitor stops is a marketing and outbound logistics problem inside the value chain, and payment choice sits in the same place.

Chargebacks, screening and false declines — disputes are a service and operations cost, and the screening that reduces them turns paying customers away, so the same activity has a revenue side.

Compliance across jurisdictions — consumer protection duties, tax obligations and intellectual property all sit in firm infrastructure, and that cost grows with every place the firm now reaches.

Showrooming — price transparency turns a trade counter into a display case for a cheaper seller, which is buyer power acting on a firm that has already paid for the building.

Working the evidence yourself

Naming the mapping is the point, and the naming has to start from something counted. That is the step readers skip, and the only one that produces facts somebody else can check. The sheet below holds the four channels the supplier sells through.

SPREADSHEET**Turn raw channel counts into evidence**

Type one formula for row 2 in each column and run it; it is applied down every row the way a filled-down formula behaves in a real sheet. The channels and figures belong to the hypothetical supplier and are invented.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Columns: Channel · Visitors · Orders · Average order value · Disputed orders · Conversion rate · Revenue · Look at this channel?

1. Column F: what share of the people who arrived actually ordered? Give it as a percentage, rounded to one decimal place.

```
=ROUND ( C2/B2*100,1)
```

A count of orders says nothing on its own, because the channels are not the same size. The ratio is what lets you compare them, and it is the first number an analysis of checkout friction has to put on the page.

2. Column G: work out what each channel actually brought in, by multiplying the orders by the average order value.

```
=C2*D2
```

Now the two columns disagree, and the disagreement is the finding. The marketplace takes more than twice as many orders as the website, and the account book takes the fewest orders of any channel here, yet the account book brings in more money than both new channels together.

3. Column H: mark any channel with more than five disputed orders for a closer look, and mark the rest as clear.

```
=IF (E2>5,"Look","Clear")
```

A flag is a decision rule written down, which is the honest way to use a threshold. Notice that it selects a channel to examine rather than a customer to refuse: the same logic applied to individual orders is what produces false declines.

A conversion rate and a revenue figure tell you where the money is. They do not tell you what the marketplace listing leaves once its commission is taken, or where the disputes are concentrated. Those are questions for the order records, and this chapter gives you every reason to ask them.

DATABASE**Ask the order records the chapter's questions**

Write a SELECT for each question and run it; any query returning the right answer is accepted. These eleven rows are one week pulled from the order file, not the year the spreadsheet totals, so the counts are small and the ratios will not match it. The supplier, its orders and its fees are hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

orders — id, channel, item, total, disputed (11 rows)

channels — channel, fee_percent, owns_customer (3 rows)

1. List the item and the total for every order that came through the marketplace listing.

```
SELECT item, total FROM orders WHERE channel = 'marketplace'
```

Choosing columns and choosing rows are separate decisions. Read the result and the pattern is already visible: the marketplace carries small, light goods, which is exactly what an entrant can post cheaply.

2. How much revenue did each channel produce in this week? Show the largest first.

```
SELECT channel, SUM(total) AS revenue FROM orders GROUP BY channel ORDER BY revenue DESC
```

Grouping turns a list of orders into a description of the channels. The account book carries the fewest orders and the most money, and the marketplace does the reverse, which is the first hint that order counts are a poor way to judge a channel.

3. Which channels carried more than one disputed order? Show the channel and how many disputes it carried.

```
SELECT channel, COUNT(*) AS disputes FROM orders WHERE disputed = 'yes' GROUP BY channel HAVING COUNT(*) > 1
```

WHERE filters rows and HAVING filters groups, because at the WHERE stage nothing has been counted yet. The answer also tells you where any screening effort should be aimed, and where it would only cost you good customers.

4. What does each channel leave the supplier after its fee? Show the channel, the gross total and the amount left once the fee is taken, best first.

```
SELECT o.channel, SUM(o.total) AS gross, SUM(o.total - o.total * c.fee_percent / 100) AS net FROM orders o JOIN channels c ON o.channel =
```

c.channel GROUP BY o.channel ORDER BY net DESC

This answer exists in neither table. The order totals sit on one and the fee sits on the other, and only the join shows the marketplace handing back fifteen percent of every order while the firm's own channels hand back under three. In this week it grossed fifty more than the website and kept five. The same table records who keeps the customer relationship, which is the supplier-power half of the story.

The method, in four steps

The analysis itself is short. Almost all of the difficulty lies in doing each step properly rather than in knowing what the steps are.

- 1 Gather the evidence.** Collect what can be counted: prices, visits, orders, disputes, the fee each channel takes, the share of revenue each one carries.
- 2 Name the pressure.** Say which force is acting and on which side of the firm. The framework earns its keep here or nowhere.
- 3 Locate the fix.** Put the pressure somewhere in the value chain, because that is what turns a diagnosis into a place where work can actually be done.
- 4 Recommend one thing.** Say what it changes, what it costs, and what you would watch afterwards to know whether it worked.

What separates a strong version of each step from a weak one is small and repeatable, which is why the two are worth seeing side by side.

STEP	WEAK VERSION	STRONGER VERSION
Gather	"Online competition is intense."	The delivered price of one comparable case of gloves, set beside your own list price.
Name	"Customers are leaving the basket."	"Buyer power is high, because comparing prices costs the customer nothing."
Locate	"We need better technology."	"The loss happens at checkout: marketing and sales, and outbound logistics."
Recommend	Three initiatives, none of them costed.	One initiative, with what it changes, what it costs and how you would tell.

DECIDE**From evidence to one recommendation**

Work the four decisions in order and read every outcome afterwards, including the ones you did not choose; the supplier is hypothetical and its figures are invented.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You are writing the analysis for the hypothetical consumables supplier. The four moments below are where this piece of work usually goes wrong, and the weaker options are the mistakes people actually make rather than obviously silly ones.

Decision 1. You have a week with the supplier's records before you write anything. What do you come back with?

- A written summary of what the counter staff and the sales reps believe is going wrong — Staff judgement is worth having and will help you read the numbers later. On its own a list of beliefs cannot be checked, and it tends to describe the day rather than locate a pressure.
- Delivered competitor prices, basket abandonment, dispute counts, channel fees and the revenue split — **the stronger call**. Right. Each of these can be counted, each attaches to a force or to a value-chain activity, and each can be checked by somebody who doubts you. Evidence is what separates an analysis from an opinion.
- A review of the marketplace storefront, noting every feature the firm's own site lacks — This produces a feature list, and a feature list quietly becomes the recommendation. You end up proposing technology because a rival has it rather than because a force demands it.
- The whole order file exported, so that nothing at all is left out of the analysis — Completeness is not evidence. Without a question in mind you cannot tell which columns matter, and the work stalls in the data rather than in the argument you were asked for.

Decision 2. Your first draft says: visitors leave the site before paying, and the maker now sells the same cases direct. What is missing?

- Nothing. It states two observed facts, and an analysis should stay close to its evidence — Staying close to the evidence is right, but an analysis is more than the evidence. A description that names no pressure gives the reader nowhere to go and no way to test whether you are correct.
- A stronger opening claim, such as calling this an existential threat to the firm — Force of language is not force in the framework sense. Escalating the wording adds urgency without adding a diagnosis, and it makes the whole piece harder to trust.
- More figures, so add the abandonment share and the dispute count to that sentence — More numbers in the same sentence still leave it a description. The figures belong in the evidence; the sentence exists to say which pressure they are evidence of.

- It names no force. Say which pressure each of those two observations is evidence of — **the stronger call**. Right. As drafted the sentence describes the day. Naming buyer power and disintermediation, and placing abandonment inside the value chain, is what lets a reader argue with you and what makes a recommendation follow from something.

Decision 3. You have three candidate initiatives you like, and the brief asks for one. What do you write?

- All three, ranked in order, so the owners can pick whichever suits the budget — Ranking looks generous and quietly hands the decision back to the reader. It also hides which pressure you were answering, and no single option ever gets costed properly.
- The most ambitious of the three, since a bolder call shows a firmer grasp — Ambition is not a criterion. The recommendation has to answer the pressure the evidence showed, and a bold initiative aimed at the wrong force is worse than a modest one aimed at the right one.
- One initiative, with the force it answers, what it costs, and what you would watch — **the stronger call**. Right. A single recommendation can be costed, argued and later judged. The other two can be named in a sentence as deliberately not chosen, which is itself an analytical move rather than a hedge.
- A new customer application, because the marketplace has one and the firm does not — This is technology with no force behind it. If you cannot say which pressure the application relieves and where it sits in the value chain, the reader has no reason to accept it.

Decision 4. The last piece is a comparison of 150 to 200 words against a company you choose. Which draft does the work?

- Name what that company does differently, then what the firm should do or refuse — **the stronger call**. Right. The comparison exists to produce a lesson for the firm you are advising. Naming the difference in the vocabulary of the frameworks is what makes it a lesson rather than a story about somebody else.
- A short profile of the company, its history, its scale and its best-known work — This is the commonest failure of the whole piece. It reads well and says nothing about the supplier, and by the end the reader knows more about a company nobody asked you to advise.
- A list of every difference between the two firms, so nothing relevant is left out — Length is not coverage. Most differences do not bear on the pressure you diagnosed, and burying the one that matters among twenty that do not is how a reader misses your point.
- An argument that the firm should copy the company as closely as its budget allows — Copying ignores what your own evidence said about this firm, which holds ground the comparison company may not hold at all. A lesson is not an instruction to imitate.

The comparison summary

The last piece is a short written comparison, 150 to 200 words, against a company you choose. It is not a profile of that company. It is a lesson drawn back to the firm you are advising.

Three things make it work: pick a comparison with a real similarity or a real contrast; name what that company does differently in the vocabulary of the frameworks; and end on what your firm should therefore do, or deliberately refuse to do.

A summary that spends its 200 words admiring somebody else has answered a different question. Every sentence should be recoverable as advice to the firm in front of you.

READY?

Could you write the analysis now?

Rate each statement honestly; anything you cannot do yet has a pointer back to the part of this module to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can name which force a described symptom belongs to, instead of repeating the symptom in different words. Go back to the list of what this chapter supplies to the frameworks, just after the mini case.
- I can explain why selling through a marketplace is a supplier-power question and not merely a channel choice. Go back to the second item in that same list, and to the fee and ownership columns in the database exercise.
- I can place cart abandonment, checkout friction and payment choice in the right part of the value chain. Go back to the mapping list, and to the second decision in the mini case.
- I can say what a chargeback costs a merchant and why a harder screening threshold has a cost of its own. Go back to the section of this module on card-not-present risk, and to the third decision in the mini case.
- I can turn raw channel counts into the two or three ratios an analysis actually argues from. Go back to the spreadsheet exercise under working the evidence yourself.
- I can state one recommendation with what it changes, what it costs, and how I would know whether it worked. Go back to the four-step method and to the third decision in the walkthrough that follows it.

REFERENCE

Chapter vocabulary

Every term the chapter defines, written so it makes sense on its own, with an example beside it. Search by word, or filter by the objective the term belongs to. Watch for the pairs that sound alike and are not: a payment service and a cryptocurrency, a

chargeback and a refund, the card security code and the card number, a copyright and a trademark.

authorization

4.5

The check a card issuer runs before a sale is allowed to go through, confirming that the card has not been reported lost or stolen and that credit is available. It is easily mistaken for proof of identity, which it is not: an approval says the card is usable, not that the person using it owns it.

For example: A stolen card number can be approved in a second and still produce a disputed charge weeks later, because nothing in the approval asked who was typing.

blockchain

4.5

An indelible, decentralized public ledger to which transactions are added in groups called blocks, each block standing as proof of everything recorded before it. Because copies live on a large number of computers, changing an old entry would need agreement from a majority of them, and entries once written are not deleted.

For example: Organizations well outside payments, from technology firms to banks, have tested the same structure for records that several parties must trust without trusting each other.

business-to-business (B2B) electronic commerce

4.1

Electronic trade between organizations rather than with an end consumer, covering suppliers, manufacturers, distributors and wholesalers. Measured by revenue it is by far the largest form of electronic commerce, and most consumer-facing firms take part in it as buyers even though shoppers never see that side.

For example: A clothing retailer buys fabric, packaging and freight capacity through supplier systems long before a single shirt appears on its own website.

business-to-consumer (B2C) electronic commerce

4.1

Electronic trade between a business and the individual people who use what it sells. It covers online retailing, digital products and streaming services, and it includes the events leading up to a purchase and the customer service that follows it.

For example: Buying a book from a large online retailer, together with the recommendations that led to it and the return that may follow.

card-not-present transaction

4.5

A sale in which no physical card is handed over, so there is no imprint of the card and no cardholder signature. The merchant carries the risk if the charge is later disputed, and has to judge from data alone whether the person paying is the cardholder.

For example: An order placed on a website at two in the morning, where everything the merchant knows about the buyer arrived through a form.

card security code

4.5

The three-digit number printed on the back of a payment card, used as evidence that whoever is paying has the physical card in hand. It is not held in the magnetic stripe, and card company rules forbid a merchant from storing it after the sale.

For example: A checkout page asks for those three digits and must then discard them, which is why a returning customer is asked for them again.

chargeback

4.5

The reversal of a card payment by the issuer after the cardholder disputes it. The merchant normally loses the transaction amount, the merchandise already shipped, the processing costs and a fee on top, and a high rate of them can lead its bank to raise fees or close the account.

For example: Vague shipping terms on a product page produce disputes that were never fraud at all, only a customer who expected something different.

consumer protection law

4.5

Rules requiring sellers to deal with customers fairly, transparently and honestly. They matter more online because the buyer cannot handle the goods before deciding, and they often apply according to where the customer lives rather than where the seller is based.

For example: A seller writes its refund terms to satisfy the strictest state it ships into, because that is the standard it will be measured against.

consumer-to-consumer (C2C) electronic commerce

4.1

Electronic trade between individuals who are not in business together, generally carried out on a site that introduces them and handles the payment. No firm owns the goods being sold; the operator supplies the meeting place and the rules.

For example: One person buys a piece of memorabilia from another through an auction site, and neither of them is a company.

copyright

4.5

Legal ownership of created work such as text, photographs, music or software. An online business has two duties at once: not to publish work owned by somebody else, and to settle in advance who will own material an outside contractor creates for it.

For example: A product page decorated with a photograph taken from another site is an infringement even though the file was free to download.

crowdfunding

4.1

Raising money for a project by collecting small amounts from many people through a website, in place of one loan or one investor. It was among the first fintech categories to reach the public, and it bypasses the institutions that used to decide which projects got funded.

For example: A designer collects enough small pledges to pay for a first production run without ever meeting a lender.

cryptocurrency

4.5

A virtual currency issued by no central bank, recorded on a shared public ledger and transferred directly between parties. The common description of such transfers as anonymous is not accurate: they are pseudonymous, because the addresses carry no names while every transfer stays permanently visible and addresses can often be tied to people.

For example: Bitcoin, the most widely used example, was launched around 2008 by a developer known only by the pseudonym Satoshi Nakamoto.

cryptographic hash

4.5

A short fixed-length value calculated from a block of data, which changes completely if any part of that data changes. Chaining these values together is what makes a ledger tamper-evident. It is a different job from encryption, which hides contents; a public ledger is meant to be readable by everyone.

For example: Altering one old transaction would change the value calculated over it and break every value calculated after it, which is how the alteration is noticed.

decentralized peer-to-peer network

4.5

An arrangement in which thousands of ordinary computers running the same software do the work a central operator would otherwise do. There is no head office that closes for a national holiday, and no single machine whose failure stops the system.

For example: A transfer is broadcast to the whole network and verified by whichever machines choose to check it, at any hour of any day.

deceptive practice

4.5

Selling conduct that misleads a buyer, such as false advertising or untrue claims about a product's quality, benefits or price. Laws in many countries prohibit it; in the United States such conduct is pursued at the federal level while other consumer questions are handled state by state.

For example: A listing showing a struck-through price the item was never actually sold at is misleading whether or not any buyer complains.

digital investment manager

4.1

A fintech service that builds and rebalances an investment portfolio through software rather than through a human adviser. It is one narrow service done cheaply, and increasingly the recommendations are produced with artificial intelligence.

For example: Somebody with a few hundred dollars to invest receives an allocated portfolio from an application, at a fee no human adviser could work for.

digital rights management (DRM)

4.5

A technological control that lets a publisher limit how digital media are copied, shared or played, including on which devices and how many times. Supporters say it protects sales; critics call it digital restriction management and argue that it inconveniences lawful buyers and can encourage the copying it was meant to prevent.

For example: A game that can be activated only a fixed number of times becomes unplayable to its lawful owner after a new computer, while an unauthorized copy carries no such limit.

digital signature

4.5

A value attached to a transaction proving it was authorized by whoever controls a particular address, without revealing the secret used to produce it. On a public ledger this is what establishes authenticity, because no bank is standing behind the transfer.

For example: A payment is accepted by strangers around the world because the signature could only have been produced by the holder of the sending address.

digital wallet

4.5

An application that stores payment credentials so a person can pay without retyping a card number or carrying the card itself. It stands between payer and merchant, so what the merchant receives is permission to charge rather than the card details.

For example: Paying for a cup of coffee by holding a phone near the terminal, with the card number never leaving the wallet.

digital watermark

4.5

An identifying mark buried inside a digital file, the electronic equivalent of the watermark placed on paper currency to deter counterfeiting. It does not prevent copying; it makes an unauthorized copy traceable back to the account that originally bought it.

For example: Music files bought from one store carry the purchaser's email address inside the file, invisible until somebody goes looking for it.

direct-to-consumer brand

4.1

A firm that sells its own products straight to the people who use them, with no wholesaler or retailer in between. It is the clearest commercial result of disintermediation, trading a distributor's reach for control of price, data and the customer relationship.

For example: A mattress company selling only through its own site knows exactly who bought what, which a company selling through shops does not.

disintermediation

4.1

The removal of the traditional parties in the middle — wholesalers, distributors, retailers — so that a producer reaches the end customer directly. The web made it practical at scale, and the same move later reached financial services, where fintech firms bypass banks and brokers.

For example: Airlines selling seats through their own sites rather than through travel agents is the standard example.

dynamic pricing

4.1

Setting a price that moves with demand instead of staying fixed on a published list. It captures more revenue for the seller when demand is high, and it is one of the practices customers most often name when they say a market has stopped feeling fair.

For example: The chapter describes a ticketing company using it on popular concerts, with prices climbing while a queue of buyers waits.

economic nexus

4.5

A tax connection to a place created by how much a seller sells into it rather than by having property or staff there. A 2018 United States Supreme Court decision established that crossing a threshold of sales can oblige a seller to collect that state's tax with no physical location in it. Thresholds differ by state and change, so the specifics are a question for somebody qualified in that jurisdiction.

For example: A small online seller shipping steadily into one state can cross its threshold in a quiet month and acquire a duty it never applied for.

e-government

4.1

The use of information systems to give citizens, organizations and other agencies information about public services, and to let them interact with government online. It has the shape of electronic commerce without a commercial sale at the centre of it.

For example: Filing a tax return over the internet rather than posting forms, which saves the filer time and the agency paper.

electronic commerce (EC)

4.1

Transactions carried out over digital networks, covering the exchange of goods, services and money among firms, between firms and their customers, and between customers themselves. It reaches beyond the moment of purchase to the research before it and the service after it.

For example: Its roots are older than the web: standard codes for transmitting cargo information by teletype were devised during the Berlin airlift in 1948.

electronic marketplace

4.1

The digital trading space created once buying and selling moved onto networks, in which a practically limitless range of services, features and functionality can be offered. Its reach and scalability are why an online presence became a strategic necessity for most companies rather than an option.

For example: A specialist seller working out of one small workshop can be found by buyers on other continents, which no physical shop window allows.

false decline

4.5

A legitimate order a merchant refuses because its screening treated the customer as risky. It costs the sale and often the customer, and it never shows up in a fraud figure, which is why tightening the rules can look like success while quietly losing money.

For example: A student ordering to a campus address while travelling abroad trips several risk signals and is turned away, though the card is genuinely theirs.

fintech

4.1

Innovative technology that transforms financial services, usually delivered by firms that do one thing very well rather than the full range a bank offers. It has accelerated electronic commerce by simplifying payment, financing and investment, and it reshapes what customers then expect everywhere else.

For example: The chapter's examples run from crowdfunding sites to online-only banks, mobile payment processors, peer-to-peer lenders and automated investment services.

fraud indicator

4.5

A characteristic of an order that experience associates with fraud, such as a mismatch between billing and shipping addresses, an email address of meaningless characters, or a burst of unusually large orders. These are signals published as guidance by a card network, not tests that establish anything: many of them describe very large numbers of entirely honest customers.

For example: Overnight shipping on a big order raises the score, and it is also what somebody does when a birthday is tomorrow.

fraud-screening service

4.5

An automated service that weighs an order against many variables and returns a measure of how risky it looks. It works better the more data the merchant collects at checkout, which is exactly the data collection that leads some shoppers to give up on the purchase.

For example: Adding a phone number and a billing address to a checkout improves screening and loses some buyers in the same moment.

identity fraud

4.5

The use of another person's personal or financial details to make transactions in their name. The chapter reports a cost to Americans running to tens of billions of dollars in a single year; treat that as a snapshot of one country at one time, and learn instead why online sales carry so much of the exposure.

For example: Card details lifted from one breached database are tested in small online orders somewhere else entirely.

import duty

4.5

A charge a country levies on goods arriving from abroad, usually payable by the recipient when the shipment lands. It is one reason a cross-border order can cost more than the checkout total suggested, and it sits alongside any tax the buyer's own jurisdiction charges.

For example: A parcel ordered from overseas is held at delivery until the recipient pays a charge that no line of the original order mentioned.

intellectual property

4.5

Creations the law treats as ownable, including written content, images, brand names and software. An online business has to work in two directions at once: not infringing what others own, and taking visible steps to protect what it owns before it loses revenue or brand distinctiveness.

For example: A firm that never posts a copyright notice and never registers its name finds an imitator trading under something very similar.

micropayment

4.5

A payment so small that ordinary processing fees would swallow it. Card networks charge a merchant a percentage of every sale, so a few cents cannot be collected profitably that way, which is one of the legitimate uses put forward for currencies whose transfer costs are negligible.

For example: Charging a reader a fraction of a cent for one article is impossible through a card network and conceivable without one.

mobile commerce

4.1

Buying and selling conducted from phones and tablets rather than from a desk. It is not simply a smaller screen: it changes where a purchase happens, how long a checkout can reasonably take, and which payment methods are practical.

For example: A shopper compares prices while standing in a store aisle and completes the order before leaving the building.

mobile payment service

4.5

A payment service built for paying in person from a phone or a small card reader rather than from a website. It offers an alternative to carrying cash or cards, and the transfers people now make to each other grew out of the same infrastructure.

For example: A market stall takes card payments through a reader plugged into a phone, with no till and no merchant terminal.

near-field communication (NFC)

4.5

A short-range radio technology, similar in spirit to Bluetooth, that lets a phone and a payment terminal exchange data when held close together. The stored payment method is charged without the card being handled, which is why contactless payment spread as quickly as handsets supported it.

For example: Holding a phone above a terminal for a second completes the payment, and the phone never displays a card number.

online marketplace

4.1

A site that hosts many independent sellers and brings its own audience to them. A seller trades margin and control for immediate reach, which is the opposite bargain from building a storefront and attracting every visitor alone.

For example: A new brand lists on a large marketplace to reach buyers on its first day, and pays the operator a share of every sale for that.

online-only bank

4.1

A bank that operates without branches, delivering accounts and payments entirely through software. Removing the branch network removes much of the cost that set the price of traditional banking, and it puts the whole relationship inside an application.

For example: An account opened from a phone in a few minutes, with no signature, no appointment and no building to visit.

payment service

4.5

An intermediary that holds a customer's payment details and pays merchants on their behalf, so the sensitive information is given once to one party instead of to every seller. The service keeps the card data, the purchase history and the contact details, and does not pass them to the merchant.

For example: One of the best known of these grew up alongside an online auction site and let strangers pay each other; the two firms were separated into independent businesses in 2015.

peer-to-peer lending

4.1

An arrangement in which people lend money directly to other people or to small businesses through a platform, rather than depositing it with a bank that lends it on. The platform matches the two sides and prices the risk, and the money never becomes a bank deposit.

For example: A borrower turned down by a bank is funded instead by a hundred individuals each putting up a small share of the loan.

peer-to-peer payment

4.5

A transfer of money directly between two individuals through an application, with neither of them acting as a merchant. It grew out of mobile payment services and is now routine for the small settlements people once handled in cash.

For example: Four friends split the bill for a meal on the spot, three of them sending their share to the one who paid the restaurant.

platform business model

4.1

A business that earns its money by enabling transactions between other parties rather than by owning what is sold. Its value grows with the number of participants on each side, which is why an established platform is hard for a newcomer to displace.

For example: A marketplace for handmade goods holds no stock at all and earns from the sales its buyers and sellers make to each other.

point-of-sale (POS) system

4.5

The combination of checkout computers and inventory control that captures a sale as it happens and deducts the item from stock. Because the record is made at the moment of sale, a retailer can see what is moving and reorder before the shelf empties.

For example: A bar code scan at a till both charges the customer and tells the ordering system that one more unit has left the shelf.

price matching

4.1

A policy under which a physical retailer agrees to sell at a competitor's advertised price. It is a direct answer to customers comparing online prices while standing in the store, and it protects the visit at the cost of some margin.

For example: Two large chains invite shoppers to check prices in the aisle, betting they will buy on the spot and add something else to the basket.

public ledger

4.5

A shared record of transactions that any computer may inspect and verify. Its openness is the point: with no central authority vouching for a transfer, correctness has to be something anyone can check, which is also the reason it cannot be private.

For example: Every transfer an address has ever made can be read by a stranger years later, which is what separates pseudonymous from anonymous.

radio frequency identification (RFID)

4.5

A tag that can be read wirelessly at a short distance, so items do not have to be scanned one at a time. In retail it would let a whole cart be totalled at once, but adoption is uneven and turns on whether an item is valuable enough to justify the cost of tagging it.

For example: A cart of tagged goods priced in a second as it passes a reader, which is practical for coats and not for tins of beans.

return and refund policy

4.5

The published terms on which a seller will take goods back and return the money. Many countries require the terms to be clear, understandable and prominently displayed, and a generous policy can raise revenue over time even though handling the returned goods is expensive.

For example: One large retailer accepts returns freely, absorbing high shipping and restocking costs because buyers order more when the risk is reversible.

risk score

4.5

A number a screening service assigns to an order to summarize how likely it looks to be fraudulent. It is an input to a decision rather than a verdict, and the merchant chooses the point at which an order is held, checked by telephone or refused.

For example: The same score might be approved by a seller of low-value goods and refused by one shipping expensive electronics overnight.

sales beacon

4.5

A small Bluetooth transmitter in a store that detects a nearby phone and triggers a real-time promotional offer. It brings the targeting familiar from online shopping into the aisle, and it raises the same tracking questions shoppers ask of websites.

For example: An offer for the coffee counter appears on a phone as its owner walks past the entrance.

sales tax

4.5

A tax on a sale that the seller collects from the buyer and passes to the authority. Online sales were long treated like mail order, meaning tax was collected only where the seller had substantial presence, and that rule has since changed. What is owed differs by jurisdiction and changes over time, so the specifics are a question for somebody qualified there.

For example: Early online sellers chose where to place warehouses partly for the tax position it gave most of their customers.

secure sockets layer (SSL) encryption

4.5

Protection applied to data moving between a browser and a website so that it cannot be read along the way. The padlock shown by the browser is the sign that it is in use, and it is the first thing to look for before sending anything sensitive.

For example: A card number typed into a page with no padlock crosses several unknown networks in a form anyone on the path could read.

self-checkout

4.5

A checkout the customer operates themselves, scanning and paying without a cashier. It saves the retailer labour costs and the shopper queueing time, and one variation lets items be scanned into the cart as they are picked up, leaving only the payment for the counter.

For example: A grocery shopper scans items while walking the aisles and finishes the visit by tapping a card on the way out.

shopping cart abandonment

4.5

The very common outcome in which a shopper fills a basket online and then leaves without buying. The chapter reports that more than half of online carts end this way, and the causes it lists include security concerns, impatience, a lengthy checkout and comparison shopping.

For example: Adding one more required field to a checkout form measurably raises the number of people who never finish the order.

showrooming

4.1

The practice of examining a product in person at a store and then ordering the same item online for less. The retailer pays for the building, the stock and the staff time while a different seller collects the sale, which is why physical retailers have had to find an answer to it.

For example: A shopper handles three televisions, asks a member of staff which is better, and orders the winner from a phone in the car park.

social commerce

4.1

Buying and selling that happens inside social platforms, where the recommendation, the discussion and the purchase sit in one place. It is one of the directions electronic commerce spread into once a transaction no longer needed a dedicated store around it.

For example: A product is discovered in a friend's post and bought without ever leaving the application it was seen in.

storefront platform

4.1

A hosted service that lets a business build and run its own online store without writing the software. It gives the seller control of branding, pricing and customer data, and it leaves the hardest part — persuading anybody to visit — with the seller.

For example: A small maker has a working store online in an afternoon, and then discovers that nobody yet knows the address.

substantial presence

4.5

The older test for whether a seller owed a state's sales tax: an office, a shipping warehouse or some other physical footing in that state. It is worth knowing because it explains where early online sellers chose to build, and because economic nexus has since added a second route to the same obligation.

For example: A seller with a single warehouse collected tax from customers in that state and, for years, from nowhere else.

third-party fulfillment

4.1

Paying an outside company to hold stock, pick orders, ship them and process returns. It converts a large investment in warehousing into a cost per order, which is how small sellers offer delivery speeds they could not build for themselves.

For example: A brand working out of a rented unit ships next day nationwide because somebody else's warehouse network is holding its stock.

trademark

4.5

A name, logo or mark identifying whose goods or services something is. Online it is easy to infringe one without meaning to, including by placing somebody else's protected name inside a domain name or a product listing.

For example: A reseller registers a web address containing a manufacturer's brand name and receives a legal letter rather than the traffic it hoped for.

use tax

4.5

A tax a buyer may owe their own state on something bought elsewhere and brought in, when no sales tax was collected at the point of purchase. It exists so that buying across a border does not simply remove the tax, and it is widely unknown to the people who owe it.

For example: Someone ordering an item from another country may owe their home state a tax that no part of the checkout mentioned.

PUT IT TOGETHER

Final challenge

Situations rather than definitions. Each one puts you in front of a decision somebody actually has to make — ship it or hold it, absorb the fee or pass it on, comply or withdraw — and asks what is really going on. Every company described is invented. Answer all of them, then score: the result breaks out by objective, so a weak area points at a section to reread instead of disappearing into one number.

FINAL CHALLENGE

Final challenge

Answer key — the interactive, self-scoring version loads when scripting is available.

1. **4.1** A hypothetical instrument maker, Kestrel Instruments, takes every final order over the telephone. Its website carries a configurator that helps a buyer specify a machine, and a portal where owners book repairs and download manuals. The sales director says the firm therefore does no electronic commerce at all. What does the chapter say?

- **A.** He is right, because the money changes hands on the phone, and electronic commerce is defined by where the payment is taken — The chapter defines electronic commerce as electronic transactions over digital networks, and it extends the term past the moment of payment to what surrounds that moment.
- **B.** He is wrong: electronic commerce covers the events leading up to a purchase and the service after it, not the checkout alone — **correct.** The chapter says electronic commerce goes beyond consumers merely buying and selling products online: it can involve the events leading up to the purchase of a product as well as customer service after the sale.
- **C.** He is right about the configurator but wrong about the repair portal, since only after-sales contact is counted as electronic commerce — Both halves belong. Helping a buyer specify a machine before the order and supporting the owner afterwards are each named by the chapter as part of electronic commerce.
- **D.** He is wrong, because any firm that operates a website of its own is by definition trading in an electronic marketplace, whatever the site does — Owning a website is not the test. The electronic marketplace is the chapter's name for the space of services and functionality that trading online opens up, not a label earned by publishing a page.

2. **4.1** A hypothetical seed merchant, Marbury Seedhouse, moves from a printed catalog to a website. Within a year it also offers sowing calendars, germination advice by crop, saved order histories, and a subscription that ships at the right week. Its founder asks what the chapter calls the space that moving online opened up, in which all of that could be offered.

- **A.** Disintermediation, because the printed catalog that used to carry the range has been removed from the chain of supply — Disintermediation is the removal of an intermediary such as a wholesaler or retailer. A catalog is a sales channel the merchant owned, not a middleman standing between it and its buyers.
- **B.** The platform business model, because the site now enables transactions between the merchant and each of its customers — A platform business model enables transactions between other parties, as an auction site does between two people. Here the merchant is itself one of the two parties in every sale.
- **C.** The electronic marketplace, in which a virtually limitless array of services, features and functionality can be offered — **correct.** The chapter says electronic commerce has led to the creation of an electronic marketplace where a virtually limitless array of services, features, and functionality can be offered. The term names that shared trading space rather than any one merchant's site, and the seedhouse has moved into it.
- **D.** Business-to-business commerce, because a subscription that ships to a schedule is a supply arrangement rather than an ordinary retail sale — Business-to-business commerce is trade among businesses, such as sourcing materials from suppliers. A grower who subscribes for their own plot is still an end consumer.

3. 4.1 A hypothetical bearing manufacturer, Cordwell Bearings, has always sold through industrial distributors. It is about to sell direct from its own site and keep the distributor margin for itself. Its operations manager warns the board that the margin does not arrive on its own. What is she pointing at?

- **A.** That cutting out the distributor also moves the distributor's work — finding buyers, taking orders, handling returns — onto the maker — **correct**. Disintermediation is cutting out the middleman and reaching customers directly; the chapter's examples are airlines selling to travelers and the rise of direct-to-consumer brands. The margin comes attached to the tasks the distributor was performing.
- **B.** That selling direct is really re-intermediation, so a new intermediary will appear and take a similar share of every order — Re-intermediation describes a new intermediary entering, such as a marketplace. Nothing here forces one to appear, because the maker is choosing to reach buyers itself.
- **C.** That the distributor's margin was never truly available, since a maker selling direct is obliged to hold the distributor's prices — Nothing in the chapter obliges a direct seller to hold the old price. The warning is about work and cost, not about what the maker is permitted to charge.
- **D.** That an online presence is optional for a manufacturer, so the safer course is to leave the existing distribution arrangement exactly as it is — The chapter argues the opposite. Because of the potential for increased market reach and scalability, an online presence has become a strategic necessity for most companies.

4. 4.1 A hypothetical ceramics workshop, Halloway Ceramics, stopped selling through galleries and now sells from its own site, but almost nobody finds it. It lists on a large online marketplace, sales recover, and the marketplace takes a share of every order. What has happened, in the chapter's terms?

- **A.** The workshop has enlarged its own electronic marketplace, because a marketplace listing adds services and features to what the workshop can offer buyers — The electronic marketplace is the chapter's term for the space of services and functionality that trading online opens up, not an asset a seller owns and enlarges by adding a channel.
- **B.** The workshop has moved from business-to-consumer selling into consumer-to-consumer selling, because a marketplace mainly hosts individual sellers — Consumer-to-consumer describes transactions between people not necessarily working together. A workshop selling its own output is a business on one side of every sale it makes.
- **C.** The workshop is now conducting business-to-business trade, because its selling agreement is with the marketplace operator rather than with the buyer — The sale that matters is still made to an end consumer. Paying a fee to a marketplace does not convert a retail transaction into trade between two businesses.
- **D.** It cut out one intermediary and took on another: the marketplace runs a platform business model and now sits between it and its buyers — **correct**. Disintermediation removed the gallery, and the marketplace is a new intermediary. The chapter notes that online marketplaces such as auction and craft sites use a platform business model, enabling transactions between other parties.

5. 4.1 A hypothetical trade body is briefing new members on where the money in electronic commerce actually sits. One member argues that consumer retail must be the largest form of it, because consumer retail is the part of it everyone sees. What does the chapter report?

- **A.** Consumer retail is indeed the largest, and trade between businesses is a small specialized corner of electronic commerce — The chapter puts it the other way round, and notes that many firms concentrate solely on trade with other businesses rather than on consumers at all.
- **B.** The two are roughly equal in size, because almost every consumer-facing firm also buys from its own suppliers online — Consumer-facing firms do buy from suppliers online, which the chapter mentions, but that observation does not make the two halves equal in revenue.
- **C.** Business-to-business trade is by far the largest form by revenue, on the manufacturing and wholesale figures the chapter reports — **correct.** The chapter says business-to-business commerce is by far the largest form in terms of revenues, citing manufacturing shipment and wholesale sales figures for a single reported year.
- **D.** Consumer-to-consumer trade is the largest, because auction and resale sites carry far more separate transactions than ordinary retailers do — Counting transactions is not the same as counting revenue. Trade between consumers involves many small sales, and the chapter does not present it as the largest by value.

6. 4.1 Two hypothetical collectors, neither of them a business, agree the sale of a vintage concert poster through an online auction site. The site handles the listing, the messaging and its own fee, but it never owns the poster. How does the chapter classify the sale, and the site?

- **A.** A consumer-to-consumer sale, with the site running a platform business model that enables the transaction between the two people — **correct.** The chapter defines consumer-to-consumer commerce as transactions between people not necessarily working together, and notes that online marketplaces such as auction and craft sites use a platform business model.
- **B.** A business-to-consumer sale, with the site as the seller, because the buyer's contract is formed on the site's own pages — The site is not a party to this sale. Business-to-consumer describes a transaction between a business and its own customer, which is not what a listing fee creates.
- **C.** A business-to-business sale, because both collectors are acting commercially the moment money changes hands for goods — Business-to-business commerce is trade among businesses, such as a manufacturer dealing with its suppliers. Two private collectors do not become firms by exchanging money once.
- **D.** Not electronic commerce at all, since the chapter restricts the term to transactions in which at least one party is a firm — The chapter's definition covers transactions among firms, between firms and their customers, and between customers, so a sale between two people is squarely inside it.

7. 4.1 A hypothetical county puts three services online in the same year: an application for identity cards, a portal publishing its spending data, and a form that lets builders file plans. A councillor asks whether any of this counts as electronic commerce.

- **A.** None of it does, because no goods change hands and the county is not selling anything to anybody at any stage of these three services — The chapter treats this as a related form that does not involve goods or services sold by or to companies, and it still gives that form a name of its own.
- **B.** All of it is business-to-consumer commerce, because a resident who applies for an identity card is in every sense a customer of the county — The chapter keeps government services separate from business-to-consumer commerce, which describes transactions between businesses and their own customers.
- **C.** Only the plans form counts, because builders are businesses, and the two services aimed at individual residents fall outside the chapter's definitions — The chapter's examples include online applications for identity cards, municipal services, electronic voting, and electronic access to public government data.
- **D.** It is e-government: information systems giving citizens, organizations and agencies information about public services and letting them interact with government — **correct.** E-government is the chapter's term for using information systems to provide citizens, organizations and other governmental agencies with information about public services and to allow interaction with the government.

8. **4.1** A hypothetical firm, Camlet Credit, does one thing: it advances money against unpaid invoices for small manufacturers, entirely online, and offers nothing else. A bank manager says it cannot be a serious financial business, because it has no current accounts, no cards and no branches. What does the chapter's account of fintech say?

- **A.** The bank manager is right, because the chapter treats fintech firms as suppliers of technology to banks rather than as providers of financial services themselves — The chapter names online-only banks, mobile payment processors, peer-to-peer lenders, small-business lenders and digital investment managers as fintech firms serving customers directly.
- **B.** Focusing on one service and doing it very well is the pattern the chapter describes, and reaching borrowers directly bypasses the traditional financial intermediary — **correct.** The chapter says that whereas traditional banks provide a wide range of services, fintech companies typically focus on one service and do this very well, and that fintechs can bypass traditional financial intermediaries to reach consumers directly.
- **C.** The firm is not doing fintech but electronic commerce, because lending money is a service sold to a customer like any other product — The chapter places online financial transactions such as banking, brokerage, lending and investment inside electronic commerce, so the two labels are not alternatives.
- **D.** Breadth is the test the chapter applies, so a firm has to match a bank's range of products before its technology is described as fintech — Range is exactly what the chapter contrasts fintech against. Narrowness is presented there as the working method, not as a disqualification.

9. 4.1 A hypothetical building society has accepted that its members now judge a two-working-day transfer against the payment apps they use for everything else. The board has to decide what follows from that, and it is split four ways. Which response fits the chapter's account of what fintech does to expectations?

- **A.** Benchmark the transfer against every other society in the region, and act only if the society turns out to be slower than its peers — Peer benchmarking answers a different question. This standard was set outside the industry, so the society can sit comfortably mid-table among societies and still lose members who have stopped comparing it with them.
- **B.** Publish an explanation of why two days is normal for a transfer of this kind, on the ground that an informed member expects less — Explaining the mechanism does not undo the comparison. The chapter treats the reshaped expectation as a fact about customers rather than a misunderstanding a firm can talk them out of.
- **C.** Treat the expectation as the standard the society is now held to, and decide whether to meet it or to compete where the apps are not — **correct.** The chapter says fintech services can greatly enhance customer experience yet also shape customer expectations, for instance in terms of ease of payment or speed. The expectation is therefore the standard the society is measured against, and the decision left is whether to meet it or to be better somewhere else.
- **D.** Wait, since an expectation formed this quickly is likely to move again before the society could finish rebuilding anything at all — Nothing in the chapter suggests these expectations recede. They are described as an effect of services members already use daily, so waiting spends the interval and changes none of it.

10. 4.5 A hypothetical tool shop, Pinfold Tools, adds four extra fields and a telephone confirmation step to its checkout after a run of fraud. Fraud losses fall. Completed orders fall further, and most of the abandoned carts belonged to ordinary customers. What does the chapter's treatment of this trade suggest?

- **A.** Security has to be maximized whatever the effect on completion, because a fraudulent order costs the shop more than a lost order does — The chapter frames this as a balance rather than a maximum, and reports that more than half of online shopping carts are abandoned even before a step is added.
- **B.** The abandoned carts are unrelated to the checkout change, since the chapter attributes cart abandonment mainly to comparison shopping — Comparison shopping is one cause the chapter lists, along with security concerns, impatience and lengthy checkout procedures — which is precisely what has just been added here.
- **C.** Telephone confirmation is the chapter's recommended default for every order, so a lower completion rate is simply the price of doing the job properly — The chapter describes calling a customer as something a merchant may do when the indicators warrant it, and notes that such calls can be problematic for privacy reasons.
- **D.** The shop should balance security against friction: collecting more data screens better, and it also drives shoppers away before they finish — **correct.** The chapter says it is essential to strike a balance between providing security and reducing friction, and notes that collecting as much data as possible during checkout may lead some customers to abandon their carts.

11. 4.5 A hypothetical map publisher, Peregrine Maps, is about to sell downloads from its own site. Its bookkeeper wants to store customer card numbers in the orders table so that repeat buyers do not have to retype them. A developer proposes accepting an established payment service instead. What is the substantive difference?

- **A.** There is none of substance: either way the publisher ends up holding the same payment details, and the service merely formats them for it — The chapter's point is about where the information comes to rest. The customer has to provide the sensitive payment information only to the payment service.
- **B.** The service settles every dispute on the publisher's behalf, which is the whole of the difference, since the card details are held in exactly the same way under both arrangements — Payment services do sit between the parties, but the benefit the chapter states here is that the service keeps the information secure and does not share it with the merchant.
- **C.** With a payment service the buyer gives sensitive details to the service alone, so the publisher never receives the card data it would otherwise have to protect — **correct.** Rather than paying a seller by providing card information, the shopper simply pays using their account with the payment service, which keeps that information and does not share it with the online merchant.
- **D.** A payment service is for transfers between people, so a publisher selling downloads would gain nothing at all by routing its sales through one — One service the chapter describes also lets anyone with an email address send and receive money, but these same services are how a great many merchants take payment.

12. 4.5 Four hypothetical colleagues share a meal. One of them pays the whole bill by holding her phone against the restaurant's terminal, and the other three send her their shares from their own phones before they leave the table. Which description matches the chapter?

- **A.** Both movements of money are peer-to-peer payments, because a phone was used on both sides of each of them — The restaurant is a business taking payment at its own terminal, which the chapter treats as an in-store mobile payment rather than a transfer between two individuals.
- **B.** The bill was a mobile payment and the three reimbursements were peer-to-peer payments, carried by the same kind of service — **correct.** The chapter describes mobile payment services facilitating in-store payments as an alternative to carrying cash or cards, and a recent development allowing peer-to-peer payments so that friends can split the bill for a meal.
- **C.** Neither is a payment service transaction, because no card number was typed into a website at any point in the evening — The chapter's payment services cover in-store taps and phone-to-phone transfers; typing a card number into a web form is not what defines them.
- **D.** The bill was a peer-to-peer payment and the reimbursements were business-to-consumer commerce settled inside a messaging app — This reverses the two. Money going to the restaurant is the business transaction, and money moving between the diners is the transfer between individuals.

13. 4.5 A hypothetical furniture importer, Wexford Timber, already accepts a cryptocurrency to avoid card fees. Buyers ask what happens when goods arrive wrong, and the importer answers with a promise of its own: it will send the coins back whenever a buyer complains. Its finance lead thinks the promise closes the gap. What does it still leave the buyer without?

- **A.** Nothing at all, because a promise written by the seller does the same work for a buyer that a card company's guarantee does — A promise the seller writes binds only the seller, and the buyer is left trusting the party they are in dispute with. That is the gap, not the wording of the promise.
- **B.** Any route back to the money, since an entry on the ledger cannot be deleted once the network has agreed to it — The coins can be sent back as a fresh transfer, so a refund is perfectly possible here. What cannot happen is a reversal imposed on a seller who has decided not to pay.
- **C.** Somebody other than the seller to settle the dispute, which is the part of the safety net the fee was buying — **correct.** The chapter says card companies and payment services act as trusted middlemen and provide consumers a safety net, giving confidence that a purchase will produce the desired good. A guarantee the seller issues and judges itself is not that.
- **D.** A record of the payment, which is why a buyer paying this way has nothing to point at when a complaint is refused — The ledger holds every transfer permanently and publicly, so evidence that the money moved is the one thing this buyer certainly does have.

14. 4.5 A hypothetical dealer in rare records tells a colleague that payments made in a cryptocurrency are anonymous and encrypted, so nobody can see who paid whom. The colleague looks up the ledger and finds every transfer listed there, going back years. Which correction is accurate?

- **A.** The ledger is encrypted, and what the colleague found was a separate public summary rather than the transactions themselves — The chapter describes a secure public ledger that is accessible to any computer wanting to verify it, so openness here is the design rather than a leak.
- **B.** Transfers are pseudonymous rather than anonymous: the ledger is meant to be readable, and hashing and digital signatures are what secure it — **correct.** Addresses carry no names, but every transfer stays permanently visible and addresses can often be tied back to people, so pseudonymous is the accurate word. Authenticity comes from digital signatures and from the hashing that chains the blocks together, not from hiding the data.
- **C.** The ledger is anonymous but temporary, because entries are cleared once the majority of nodes have confirmed the block that holds them — Entries are not cleared. The chapter calls the ledger indelible and says the chain serves as proof of all transactions ever made within the network.
- **D.** The dealer is right, and the colleague was reading a test network, since a live ledger is visible only to the computers that maintain it — There is one shared public ledger in the chapter's account, constantly verified and maintained by the network and open every day of the year.

15. 4.5 A hypothetical audio brand, Quillon Audio, is told that because a cryptocurrency's network runs reliably and its fees are negligible, customers will use it as soon as the option appears at checkout. The finance lead is unconvinced. What does the chapter's record of merchant trials suggest?

- **A.** Adoption follows capability, so the option should be added and take-up will arrive along with it in due course — The chapter's account runs the other way: the capability existed, was offered by well-known sellers, and the buying public did not follow it.
- **B.** Merchants abandoned it because the network could not settle payments quickly enough at the till, which is a technical fault rather than a question of what buyers are used to — The chapter attributes the outcome to a lack of use rather than to a settlement failure, while noting that interest in the currency itself continued.
- **C.** The chapter reports steady growth in purchases made this way, so the finance lead is arguing against the trend the evidence describes — The chapter records continued interest in the currency, but it describes use for making purchases as negligible rather than as growing steadily.
- **D.** Several large sellers accepted it and later stopped, and purchase use stayed negligible, so a working mechanism is not an adopted one — **correct.** The chapter reports that several major online companies announced they would accept the currency, that those trials have been terminated, and that its use for making purchase transactions has remained negligible.

16. 4.5 A hypothetical kayak maker, Duskwater Kayaks, receives an approved authorization on a large order and ships the same hour, telling staff that the approval confirms the buyer is the cardholder. Two weeks later the real cardholder disputes the charge. What did the approval actually establish, and what was thrown away by shipping at once?

- **A.** It established the buyer's identity, and nothing was thrown away, because a disputed transaction becomes the issuing bank's loss once that bank has approved it — The chapter says the merchant is financially responsible for disputed transactions, and that card issuers typically charge back transactions disputed by cardholders.
- **B.** It established only that the card was not reported lost or stolen, and shipping at once used up the days before dispatch that could have gone on verification — **correct.** Authorization merely ensures the card was not reported lost or stolen and does not ensure the person making the transaction is the actual cardholder. Because the transaction date is the date the merchandise ships, merchants typically have one or several days to verify identity — a window that does not exist for a digital product delivered at once.
- **C.** It established that the card was in the buyer's hands, and shipping at once made no difference, because the window runs from the order date — Possession is what the security code speaks to rather than the authorization, and the chapter ties the available days to the shipping date, not to the order date.
- **D.** It established that the billing and shipping addresses agree, and the maker gave up its chance to charge a higher price for overnight delivery — Address agreement is one of the variables a screening service weighs. It is checked by the merchant rather than established by the issuer's approval.

17. 4.5 A hypothetical lighting supplier, Ombra Lighting, wants to keep the three-digit code from the back of each customer's card in its own customer file, so that returning buyers can skip a step and so that disputes can be re-checked later. What is wrong with the plan?

- **A.** Nothing, provided the file is encrypted, since the code is printed on the card and is not sensitive information in the chapter's account — The chapter describes the code as a mechanism used to authorize card-not-present transactions, and it records an explicit prohibition on merchants keeping it.
- **B.** The code sits in the card's magnetic stripe already, so keeping a second copy adds risk without adding anything the supplier does not have — The chapter says the opposite: the code is not stored in the card's magnetic stripe, which is exactly why possession of the card is what it demonstrates.
- **C.** Card company rules prohibit merchants from storing it, and storing it destroys what it proves: that someone had the physical card in hand — **correct.** The chapter notes that credit card company rules prohibit merchants from storing the code, and that it is known only to a person who is in possession of the physical card.
- **D.** The code changes with every transaction, so a stored copy would be stale before the next order and could not be re-checked against anything — A printed code does not rotate between orders. Its value comes from where it is printed, and the rule against keeping it stands in any case.

18. 4.5 A hypothetical stationery seller, Larkfield Paper, builds its first storefront. It lifts the product photographs straight from its supplier's catalogue, and registers a domain with a well-known pen brand's name inside it because that is what buyers type into a search box. Its owner says nothing is wrong, since every item listed is genuine stock the shop has paid for. Where is the difficulty?

- **A.** Neither move is a problem, because the goods are genuine and the seller can show an invoice for every item it lists — Stocking the item settles nothing about whose intellectual property appears on the page. Whether the goods are genuine is a different question from who owns the photograph or the name.
- **B.** Only the domain is a problem, since a photograph published in a public catalogue may be reused by anyone selling the item shown in it — A catalogue photograph is a work somebody made and owns, and being able to see it transfers no right to publish it. Permission is what the chapter tells a seller to obtain first.
- **C.** Both are infringements in the chapter's own examples: the photograph is copyright, and the name in the domain is a trademark — **correct.** The chapter calls ownership of content and trademarks a virtual minefield, and its two examples of infringing are exactly these: posting product photographs without permission, and having a trademarked name in your own domain name.
- **D.** Only the photographs are a problem, and the domain is safe as long as the seller states its own jurisdiction clearly on the site — Stating the jurisdiction and complying with it is separate housekeeping the chapter also asks for. It says nothing about whose mark sits inside the domain, and it makes that registration no safer.

19. 4.5 A hypothetical outdoor clothing seller, Ashgrove Outfitters, sets its system to reject any order whose shipping address differs from the billing address. Rejections rise, confirmed fraud falls a little, and the support inbox fills with people sending gifts or shipping to their workplace. How should the seller read that indicator?

- **A.** As a signal that should move a risk score alongside other variables, because a mismatch also describes a great many legitimate buyers — **correct**. Address comparison is one variable among the time of the order, the transaction volume and the customer's network location; a refused genuine order costs the sale and often the customer as well, which is why one signal should not decide alone.
- **B.** As a test that settles the question, so the rule is right and the complaints are the unavoidable cost of controlling fraud — The chapter presents address agreement as one of several variables feeding a risk score that the merchant then judges, not as a decision that stands on its own.
- **C.** As evidence that the screening service is miscalibrated, so the right response is to take address comparison out of the risk score altogether and rely on other variables — Removing the variable discards real information. The chapter includes the match between shipping address, billing address and telephone number among the useful inputs.
- **D.** As a reason to require telephone confirmation on every order, which takes the judgment out of the system and hands it to staff — Calling every customer would be slow and costly, and the chapter notes that verification calls can be problematic for privacy reasons.

20. 4.5 A hypothetical online stationer trades from one small office and ships everywhere. Its owner argues that only the rules of the place where the office sits can apply to the business, because that is the only place it has any presence at all. What is the shape of the duty the chapter describes?

- **A.** Only the seller's own location matters, which is why sellers choose a home base with the lightest consumer rules available to them — The chapter describes sellers choosing locations for tax reasons, but it treats consumer protection as attaching to the buyer rather than to the seller.
- **B.** Consumer protection rules commonly attach to where the customer is, so many sellers work to the strictest requirements they face — **correct**. The chapter says many consumer protection laws apply based on the location of the consumer, irrespective of whether a business has a presence in that location, which is why many sellers strive to comply with the strictest regulations they meet. Which rules bind a particular sale is a question to establish with somebody qualified in that jurisdiction.
- **C.** Consumer rules apply solely where the goods are physically delivered, so a download sold abroad falls outside every consumer regime — Delivery is not the hinge the chapter uses, and it describes the absence of physical interaction online as a reason these rules matter more rather than less.

- **D.** A single harmonized international standard governs online selling, so one compliant policy answers every market the stationer ships to — The chapter notes an attempt to harmonize rules within one bloc, while describing oversight elsewhere as split between a national body and state-level authorities.

21. **4.5** A hypothetical mail-order tea merchant has no office, warehouse or staff outside its own state, and has always told buyers elsewhere that no sales tax applies to them. Its sales into one neighboring state have grown steeply this year. Why might that growth change its obligations?

- **A.** Growth cannot change anything, since a duty to collect follows a physical presence and the merchant has no presence there — The chapter describes exactly this rule changing: presence is no longer the only route by which a duty to collect can arise for an online seller.
- **B.** The obligation would fall on the buyers as an import duty payable on arrival, so the merchant's own position is unchanged however fast its sales into that state grow — Import duty is the chapter's example for goods crossing a national border, not for a shipment moving between two states inside one country.
- **C.** A duty arises the moment any sale is made into another state, so the merchant has been wrong from the first order it ever shipped — The older treatment the chapter describes required a substantial presence, so the merchant's original position was not wrong at the time it was adopted.
- **D.** Crossing a sales threshold in a state can create a duty to collect there with no physical presence, which is why volume now matters — **correct.** The chapter describes economic nexus, under which states can collect sales taxes from online sellers even without a physical location once certain sales thresholds are reached, and the tracking burden that follows. What any particular threshold is now is a question to establish with somebody qualified in that jurisdiction.

22. **4.5** A hypothetical audiobook publisher is deciding whether to lock its files so that they play only on registered devices and only a set number of times. Its editor is for the idea and its head of support is against it. What is the argument on each side, as the chapter presents them?

- **A.** For: locking discourages unauthorized copying and protects sales. Against: the restrictions inconvenience buyers, and some firms succeed without them — **correct.** The chapter says the entertainment industry argues these controls minimize sales losses by preventing unauthorized duplication, while critics say publishers are arbitrary in how they enforce them, that limits on devices and activations inconvenience users and can breed piracy, and that some businesses do very well without them.
- **B.** For: locked files load faster on a registered device. Against: the encoding cost is high and rises with the size of the catalog — Loading speed is not part of the argument. The chapter frames the technology as a control on copying and distribution rather than as a performance feature.

- **C.** For: the law requires publishers to restrict copying. Against: enforcement is impossible once a file has left the publisher's servers — The chapter presents this as a commercial choice publishers make, not a legal requirement, and it names successful sellers who have chosen the other way.
- **D.** For: watermarking each file identifies the buyer. Against: watermarks reduce audio quality enough for listeners to notice the difference — Watermarking is the chapter's alternative for unlocked files, used to trace an illegal copy back to the original purchaser rather than to prevent playback.

23. **4.6** A hypothetical maker of wooden toys builds a storefront on a hosted commerce platform, and after three months has had forty visitors. Its founder concludes that the platform was badly built. What has the founder misread about the layers the supplement describes?

- **A.** Nothing at all: a storefront platform is sold as a complete solution to selling online, so weak traffic is a defect in the product the maker paid for — The supplement separates the layers precisely because no single one of them covers the whole job of selling something online.
- **B.** A storefront platform builds and runs the shop; attracting customers to it is a separate problem, which is why sellers turn to marketplaces — **correct.** Building and running your own online store still requires attracting customers to your website, which is a challenge for a fledgling business; that gap is what a marketplace with an existing customer base fills.
- **C.** The platform layer includes fulfillment, so the founder should be measuring deliveries rather than visits during a first quarter of trading — Fulfillment is a separate layer in the supplement, handled by third-party providers managing warehousing, shipping and returns rather than by the storefront software.
- **D.** Payment providers are the layer that brings demand, so the founder should change the checkout options before touching anything else — Payment providers reduce friction at checkout for buyers who have already arrived. They do not bring anyone to a shop that nobody has found.

24. **4.6** A hypothetical soap maker lists on a large marketplace and its orders multiply. A year later it wants to email those buyers about a new range, and finds that it cannot reach most of them. What was the trade it made?

- **A.** It borrowed the marketplace's audience and technology, and the marketplace kept the customer relationship that the audience produced — **correct.** A marketplace lends a seller its customer base and its technology, connecting a small seller with a very large audience at once; what it does not hand over is the relationship with the people who buy.
- **B.** It bought advertising space, so the customers were rented for the length of the campaign and returned to the marketplace afterwards — Advertising buys attention for a period. A marketplace listing is a sales channel, and the limit here is on contact afterwards rather than on the length of a campaign.
- **C.** It became a business-to-business seller, so the buyers are the marketplace's customers and the soap maker's single customer is the marketplace — The people buying soap are consumers. Paying a marketplace a fee does not turn a retail sale into trade between two businesses.

- **D.** It disintermediated its own shop, so the marketplace is now the one route by which any of its products can reach a buyer — Disintermediation means removing a middleman. The soap maker added one, and nothing stops it from continuing to sell through its own site as well.

25. **4.6** A hypothetical seller of climbing holds packs orders in a rented garage and turns away a spike of demand every autumn, because it cannot pick and post fast enough. A third-party provider quotes a per-order fee higher than the seller's own handling cost. When does that fee stop being simply an expense?

- **A.** It does not, because a fee above the seller's own cost per order is a worse deal on every single order it touches — Cost per order is not the whole comparison. The supplement points at speed, at scale, and at the orders the seller currently cannot accept at all.
- **B.** When the provider agrees to hold the stock at no charge, since warehousing is the part of fulfillment worth handing over — Warehousing is one part of the service. The providers described also handle stock, shipping and returns, and speed of delivery is a large part of the value.
- **C.** Once the seller has grown enough to build its own warehouse, at which point the fee becomes an investment in the eventual move — Building a warehouse is the significant upfront investment the arrangement is meant to avoid, so a fee paid now is not a deposit against one.
- **D.** When it stops being a comparison of fees and becomes the purchase of delivery speed and capacity the seller cannot fund itself — **correct.** Third-party providers offer rapid delivery, warehousing, shipping and returns handling, and scalable logistics without significant upfront investment — which is what a garage cannot supply at the autumn peak.

26. **4.6** A hypothetical piano dealer finds customers playing instruments in its showroom and then ordering the same model online for less. It is weighing three responses: matching online prices, offering fitting and tuning that a website cannot, and putting sensors in the aisles to track how long shoppers linger. What does the supplement say about the third one?

- **A.** Tracking is the strongest of the three, because measurement is what lets the other two responses be judged and adjusted over time — Measurement can be useful, but the supplement presents service and price matching as the answers to showrooming and treats in-store tracking as a separate question with objections attached to it.
- **B.** Price matching and personal service address the price gap directly, while tracking raises a privacy objection the dealer has to answer openly — **correct.** Retailers answer showrooming with price-matching policies and superior, personal customer service, benefiting by selling additional products, while privacy advocates criticize the tracking of purchasing habits, so the dealer has to say plainly what is collected and why.
- **C.** Tracking is equivalent to price matching, because both are ways of collecting more value from a shopper who is already inside the shop — The two do different work. One closes the price gap the shopper walked in with; the other gathers information about the shopper and needs consent to be defensible.

- **D.** Tracking is required before price matching can be offered, since a dealer has to know which shoppers were comparing before it can match anything — A price-matching policy is published and applies to anyone who asks for it. It does not depend on identifying which shoppers have been comparing prices.

27. **4.7** A hypothetical bicycle shop's written analysis opens with one line: sales are down, and customers complain about the website. The writer then moves straight on to a recommendation. Which step of the method has been skipped, and what would repair it?

- **A.** It names no numbers, so setting last year's revenue beside each complaint would repair it — Counting belongs to step one, whose stronger version is a delivered competitor price set beside your own shelf price. A number attached to a symptom is still a symptom.
- **B.** It is written in the first person, so restating the two observations in a neutral voice would repair it — Voice is a matter of style. What is missing is the pressure itself, which is what the second step of the method exists to supply.
- **C.** These are symptoms: step two asks which pressure is acting on the shop, and on what evidence — **correct.** Step two is to name the pressure and say which side of the firm it is acting on. The weak-against-stronger table sets "customers are leaving the basket" against "buyer power is high, because comparing prices costs the customer nothing", and this analysis has written only the weak version.
- **D.** It covers one force only, so a sentence about each of the other four would repair it — Four more sentences would still name no pressure a reader could argue with. The step is finished when a force is named with evidence behind it, not when five headings have been filled in.

28. **4.7** A hypothetical dairy's analysis names two areas where a system would help: the website, and the online shop. Each description runs to a single sentence, and both of them point at the same ordering screen. What is the defect?

- **A.** Two entries, one place: the chain has been located once, so nothing has been learned about the rest of it — **correct.** Step three is to locate the pressure somewhere in the value chain, and two names for one ordering screen locate it in a single place rather than in two.
- **B.** The entries are too short, so expanding each of them into a paragraph would repair it without changing what they name — Length is not the failure here. A longer description of the same ordering screen would still point at a single place in the chain.
- **C.** Neither entry is a system, so the writer should have named the software product it intends to buy for each of the two areas — Naming a product belongs to the recommendation. Locating is about where in the chain the pressure sits, which is settled before anyone decides what to buy.
- **D.** Ordering is a support activity rather than a primary one, so it sits outside the part of the chain worth locating — Support activities carry pressures too: this supplement maps compliance duties into firm infrastructure. The category of the activity is not what makes this pair weak.

29. 4.7 A hypothetical optician's analysis ends with three ideas — online booking, a loyalty scheme, and a stock system — and invites the reader to choose between them. Each idea is described in a phrase. What does the analysis need instead?

- **A.** The same three ideas ranked by cost, so that the reader can see which is cheapest to start with and choose on that basis — Ranking still hands the decision back to the reader, and the fourth step of the method asks the writer to make the call and then stand behind it.
- **B.** A fourth idea drawn from a different part of the chain, so that the reader has a wider set of options to weigh before deciding for themselves — Adding an option moves in the wrong direction. The difficulty is that no choice has been made, not that too few were listed.
- **C.** The three ideas described at greater length, since the weakness is the phrase-length treatment rather than the number of ideas offered — Length would improve each description, but three described ideas with no recommendation among them is still a menu rather than a recommendation.
- **D.** One initiative, with the pressure it answers, what it changes, what it costs, and what you would watch afterwards — **correct.** Step four is to recommend one thing and say what it changes, what it costs and what you would watch afterwards to know whether it worked. The table sets three uncoded initiatives against exactly that, and the two ideas you drop can be named as deliberately not chosen.

30. 4.7 A hypothetical bookbinder's comparison section says that a very large retailer built a network of automated warehouses, and concludes that the bookbinder should build automated warehouses too. The section runs to ninety words. Name what is wrong with it.

- **A.** The lesson only works at the larger firm's budget and footprint, and the section sits well below the length the task sets — **correct.** A lesson that only holds at the larger company's budget or footprint is not a lesson that transfers, and the comparison is set at a hundred and fifty to two hundred words, which ninety does not reach.
- **B.** The comparison is with a real company, which the task does not permit, so an invented example of a similar size would have been safer — Nothing in the method turns on whether the comparison company is real. What has to work is the lesson drawn back to the firm you are advising.
- **C.** The lesson is sound and the single defect is length, so extending the section past the lower bound would finish the job — Length is one of the two faults. A longer passage recommending warehouse automation to a bookbinder would still be recommending something out of reach.
- **D.** The company chosen is in a different industry, and a comparison has to be drawn from the same trade to carry any weight — Industry is not what decides it. A comparison fails when the lesson does not transfer, and warehouse automation does not survive the move to a bookbinder's scale and footprint.

What to do next. Buy something small online this week and watch the machinery instead of the product. Count the steps between wanting it and paying for it, and ask which of them exists to reduce your risk and which exists to reduce the seller's. Notice who holds your card details at the end. Notice what the return policy actually promises, and where the seller says its jurisdiction is. Then ask the question this module keeps returning to: if this transaction goes wrong, whose money is it, and who decides?

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